

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-08

Contents

Executive Summary	2
Headline state	2
Today's events	2
Full Sample PC1	3
Top 10 PC1 weights	3
Cluster Overview	5
Cluster Details	6
Cluster 1 – Mixed: IYY, SCHX, ILCB	6
Cluster 2 – Mixed: IJH, MDY, IVOO	9
Cluster 3 – Mixed: VEA, SPDW, SCHF	13
Cluster 4 – Mixed: BND, AGG, SCHZ	16
Cluster 5 – Mixed: VTV, SCHV, IWD	19
Cluster 6 – Mixed: IWP, VOT, IMCG	21
Cluster 7 – Emerging Markets Equity (19/26)	22
Cluster 8 – High Yield Bond (10) + Preferred Stock (7)	24
Cluster 9 – Leveraged / Inverse (16/22)	25
Cluster 10 – Healthcare (16/17)	26
Cluster 11 – Materials (13/17)	27
Cluster 12 – Financials (13/15)	28
Cluster 13 – Mixed: DBP, GLTR, NUGT	29
Cluster 14 – Mixed: INDA, INDY, EPI	30
Cluster 15 – China Equity (13/15)	31
Cluster 16 – Energy (10/14)	32
Cluster 17 – Real Estate (14/14)	33
Cluster 18 – Municipal Bond (13/13)	34
Cluster 19 – Consumer Staples (6/11)	35
Cluster 20 – Broad Commodities (6/10)	36

Cluster 21 – Inflation-Protected Fixed Income (10/10)	37
Cluster 22 – Japan Equity (7/9)	38
Cluster 23 – Mixed: SRLN, BKLN, FTSL	39
Cluster 24 – Cash / Ultra-Short Fixed Income (9/9)	40
Cluster 25 – Energy (5/8)	41
Cluster 26 – Currency (5/7)	41
Cluster 27 – Mixed: UUP, EUO, TBT	42
Diversifier Candidates	44
Long ETF Positions	45
Today's Crossovers	55
Buy crossovers	55
Sell crossovers	55
Net Signal Strength (NSS)	57
Top 20 most bullish (NSS desc)	57
Top 20 most bearish (NSS asc)	57
Run Parameters	59

Executive Summary

Run date: 2026-05-08 **Universe:** 728 ETFs across 27 clusters

Headline state

- **Full sample PC1:** LONG since 2026-04-09 (19 days, 5.7%)
- **Clusters in long state:** 15 of 27 (56%)
- **ETFs in long state:** 439 of 728 (60%)

Today's events

- Buy crossovers: 30
- Sell crossovers: 4
- Imminent buys (P > 50% within 5 days): 687
- Imminent sells (P > 50% within 5 days): 283

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with 5d / 20d / 35d MAs (last 12 months).

Current state: LONG since 2026-04-09 (19 days, 5.7%)

5dma: 144.112 20dma: 142.133 35dma: 138.172

Spread vs 35dma: +5.940 Spread vs 20dma: +1.979

P(buy crossover within 5 days): 100.0% P(sell): 4.8%

Top 10 PC1 weights

	ticker	weight	name	category
108	ACWI	0.0019	iShares MSCI ACWI ETF	Global Equity
119	AOA	0.0019	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation
122	AOR	0.0019	iShares Core 60/40 Balanced Allocation ...	Asset Allocation
126	URTH	0.0019	iShares MSCI World ETF	Global Equity
128	CRBN	0.0019	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity
139	SPGM	0.0019	State Street SPDR Portfolio MSCI Global...	Global Equity
143	VT	0.0019	Vanguard Total World Stock Index ETF	Global Equity

186	RSP	0.0019	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity
288	FEX	0.0019	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities
299	JHMM	0.0019	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity

Cluster Overview

	id	name	category	basket	n	var_explained	signal	entry_date	days	cum_return	nss
1	cluster_1	Mixed: IYY, SCHX, ILCB	Mixed	SCHX+XNTK+XSD (cor=1.00)	112	0.873	LONG	2026-04-10	18	11.4%	1.000
23	cluster_23	Mixed: SRLN, BKLN, FTSL	Mixed	SRLN+LQDH+FLOT (cor=0.97)	9	0.560	LONG	2026-04-02	23	1.2%	1.000
24	cluster_24	Cash / Ultra-Short Fixed Inco...	Cash / Ultra-Short Fi...	GSY+NEAR+VNLA (cor=0.94)	9	0.456	FLAT				1.000
25	cluster_25	Energy (5/8)	Energy	QCLN+REMX+ICLN (cor=0.99)	8	0.774	LONG	2026-04-10	18	12.9%	1.000
20	cluster_20	Broad Commodities (6/10)	Broad Commodities	GSG+UCO+DBC (cor=1.00)	10	0.912	FLAT				0.999
5	cluster_5	Mixed: VTV, SCHV, IWD	Mixed	VTV+SDY+SPYV (cor=1.00)	49	0.899	LONG	2026-04-10	18	2.9%	0.998
7	cluster_7	Emerging Markets Equity (19/2...	Emerging Markets Equi...	IEMG+DEM+FEMS (cor=1.00)	26	0.879	LONG	2026-04-10	18	10.3%	0.993
19	cluster_19	Consumer Staples (6/11)	Consumer Staples	VDC+FUTY+PSCC (cor=1.00)	11	0.748	LONG	2026-04-27	7	0.6%	0.990
6	cluster_6	Mixed: IWP, VOT, IMSG	Mixed	IWP+BLOK+ARKQ (cor=0.99)	35	0.781	LONG	2026-04-10	18	10.9%	0.986
21	cluster_21	Inflation-Protected Fixed Inc...	Inflation-Protected F...	TIP+STPZ+SPIP (cor=1.00)	10	0.908	LONG	2026-04-09	19	0.6%	0.964
16	cluster_16	Energy (10/14)	Energy	FENY+AMLX+XES (cor=1.00)	14	0.861	LONG	2026-04-29	5	-2.9%	0.919
17	cluster_17	Real Estate (14/14)	Real Estate	SCHH+RWR+INDS (cor=1.00)	14	0.962	LONG	2026-04-10	18	4.5%	0.902
2	cluster_2	Mixed: IJH, MDY, IVOO	Mixed	IJH+SPSM+IWR (cor=1.00)	107	0.889	LONG	2026-04-09	19	5.1%	0.867
26	cluster_26	Currency (5/7)	Currency	UDN+BWX+FXF (cor=0.99)	7	0.763	LONG	2026-04-13	17	0.8%	0.823
15	cluster_15	China Equity (13/15)	China Equity	CXSE+CNXA+MCHI (cor=0.99)	15	0.804	LONG	2026-04-13	17	3.7%	0.785
8	cluster_8	High Yield Bond (10) + Prefer...	High Yield Bond	JNK+PFF+PCY (cor=0.99)	22	0.772	LONG	2026-04-09	19	1.1%	0.525
22	cluster_22	Japan Equity (7/9)	Japan Equity	EWJ+DXJ+VPL (cor=1.00)	9	0.888	FLAT				0.474
3	cluster_3	Mixed: VEA, SPDW, SCHF	Mixed	VEA+VGK+VYMI (cor=1.00)	82	0.839	FLAT				0.398
12	cluster_12	Financials (13/15)	Financials	VFH+KBE+KCE (cor=1.00)	15	0.883	FLAT				0.294
27	cluster_27	Mixed: UUP, EUO, TBT	Mixed	DUST+TBT+DZZ (cor=0.98)	6	0.459	LONG	2026-05-01	3	-5.2%	0.255
18	cluster_18	Municipal Bond (13/13)	Municipal Bond	VTEB+HYD+CMF (cor=0.99)	13	0.819	FLAT				0.237
11	cluster_11	Materials (13/17)	Materials	GNR+IYM+PICK (cor=0.99)	17	0.756	FLAT				0.220
10	cluster_10	Healthcare (16/17)	Healthcare	FXH+XBI+VHT (cor=0.99)	17	0.764	FLAT				0.195
14	cluster_14	Mixed: INDA, INDY, EPI	Mixed	INDA+ILF+EIDO (cor=0.97)	15	0.386	FLAT				-0.007
4	cluster_4	Mixed: BND, AGG, SCHZ	Mixed	BND+BLV+BIV (cor=1.00)	59	0.833	FLAT				-0.347
13	cluster_13	Mixed: DBP, GLTR, NUGT	Mixed	GLTR+GDJX+AGQ (cor=1.00)	15	0.831	FLAT				-0.763
9	cluster_9	Leveraged / Inverse (16/22)	Leveraged / Inverse	SDS+UVXY+TZA (cor=1.00)	22	0.722	FLAT				-0.980

Cluster Details**Cluster 1 – Mixed: IYY, SCHX, ILCB****Members:** 112 **Var explained:** 0.873 **Signal:** LONG since 2026-04-10 (18 days, 11.4%)**Category breakdown**

category	n_members	abs_weight
Technology	23	0.2010
U.S. Large Cap Growth	11	0.1027
U.S. Large Cap Equity	9	0.0852
U.S. Total Market Equity	7	0.0664
Global Equity	7	0.0651
Consumer Discretionary	6	0.0499
Leveraged / Inverse	5	0.0471
U.S. Equity	4	0.0373
Asset Allocation	4	0.0341
Communication Services	4	0.0322
U.S. Mega Cap Equity	3	0.0282
U.S. Quality / Factor Equity	3	0.0274
U.S. Multifactor Equity	3	0.0273
Unknown	3	0.0259
U.S. ESG Equity	2	0.0189
Growth Equity	2	0.0184
Commodity	1	0.0094
Large Cap Blend Equities	1	0.0094
Large Cap Value Equities	1	0.0093
U.S. Mega Cap Growth	1	0.0093
U.S. Equal Weight Equity	1	0.0092
Dividend Equity	1	0.0091
U.S. Dividend Equity	1	0.0088
Options Income	1	0.0085
U.S. Small Cap Technology	1	0.0083
Alternatives	1	0.0082
Trend-Following Equity	1	0.0080
International IPO Equity	1	0.0074
Volatility	1	0.0072
U.S. Trend-Following Equity	1	0.0072
Israel Equity	1	0.0070
Singapore Equity	1	0.0065

Replication basket – SCHX+XNTK+XSD (cor=1.00) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
SCHX	0.820	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
XNTK	0.175	State Street SPDR NYSE Technology ETF	Technology	LONG
XSD	0.052	State Street SPDR S&P Semiconductor E...	Technology	LONG

Members

	ticker	weight	name	category	etf_signal
10	SPY	0.0095	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
70	IWV	0.0095	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
71	IYY	0.0095	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
74	SPYX	0.0095	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
76	ESGU	0.0095	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
77	IWB	0.0095	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
78	IWL	0.0095	iShares Russell Top 200 ETF	U.S. Equity	LONG
82	VTHR	0.0095	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
85	VONE	0.0095	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
86	MGC	0.0095	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
87	VV	0.0095	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
88	ILCB	0.0095	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
89	ITOT	0.0095	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
90	IVV	0.0095	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
91	SCHB	0.0095	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
92	SCHX	0.0095	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
93	SPLG	0.0095			LONG
94	SPTM	0.0095	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
95	VOO	0.0095	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
96	VTI	0.0095	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
99	SSO	0.0095	ProShares Ultra S&P500	Leveraged / Inverse	LONG
102	UPRO	0.0095	ProShares UltraPro S&P 500	Leveraged / Inverse	LONG
103	SPXL	0.0095	Direxion Daily S&P 500 Bull 3X ETF	Leveraged / Inverse	LONG
14	ACWI	0.0094	iShares MSCI ACWI ETF	Global Equity	LONG
19	URTH	0.0094	iShares MSCI World ETF	Global Equity	LONG
42	ONEQ	0.0094	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
44	IWF	0.0094	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
53	VONG	0.0094	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
55	ILCG	0.0094	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
56	IUSG	0.0094	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
68	DSI	0.0094	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
69	SUSA	0.0094	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
72	LRGF	0.0094	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
73	OEF	0.0094	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG

83	GSLC	0.0094	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
11	QQQ	0.0093	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
20	CRBN	0.0093	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
21	SPGM	0.0093	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
22	VT	0.0093	Vanguard Total World Stock Index ETF	Global Equity	LONG
41	NULG	0.0093	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
43	IWY	0.0093	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
45	IVW	0.0093	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
48	VOOG	0.0093	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
54	MGK	0.0093	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
57	SCHG	0.0093	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
58	SPYG	0.0093	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
59	VUG	0.0093	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
60	TILT	0.0093	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
66	JHML	0.0093	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
75	XLG	0.0093	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
79	QUAL	0.0093	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
84	EPS	0.0093	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
97	TQQQ	0.0093	ProShares UltraPro QQQ	Leveraged / Inverse	LONG
106	QLD	0.0093	ProShares Ultra QQQ	Leveraged / Inverse	LONG
15	AOA	0.0092	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
25	QQEW	0.0092	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
37	IGM	0.0092	iShares Expanded Tech Sector ETF	Technology	LONG
40	IYW	0.0092	iShares U.S. Technology ETF	Technology	LONG
47	VGT	0.0092	Vanguard Information Tech ETF	Technology	LONG
52	FTEC	0.0092	Fidelity MSCI Information Technology In...	Technology	LONG
63	IOO	0.0092	iShares Global 100 ETF	Global Equity	LONG
65	CATH	0.0092	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
1	XLK	0.0091	State Street Technology Select Sector S...	Technology	LONG
29	XT	0.0091	iShares Future Exponential Technologies...	Technology	LONG
36	PWB	0.0091	Invesco Large Cap Growth ETF	Growth Equity	LONG
38	IXN	0.0091	iShares Global Tech ETF	Technology	LONG
64	QDF	0.0091	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
8	AIQ	0.0090	Global X Artificial Intelligence & Tech...	Technology	LONG
18	AOR	0.0090	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
26	QTEC	0.0090	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
31	FXL	0.0090	First Trust Technology AlphaDEX	Technology	LONG
32	XNTK	0.0090	State Street SPDR NYSE Technology ETF	Technology	LONG
61	SPHB	0.0090	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
80	QUS	0.0090	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
81	SPHQ	0.0090	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
28	TDIV	0.0089	First Trust NASDAQ Technology Dividend ...	Technology	LONG
7	ROBO	0.0088	ROBO Global Robotics and Automation Ind...	Technology	LONG
39	IYC	0.0088	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
67	DGRW	0.0088	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG

46	VCR	0.0087	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
51	FDIS	0.0087	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
107	FIVG	0.0087			LONG
108	OMFL	0.0087	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
2	XLY	0.0086	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
6	BOTZ	0.0086	Global X Robotics & Artificial Intellig...	Technology	LONG
23	NXTG	0.0086	First Trust Indxx NextG ETF	Technology	LONG
35	QYLD	0.0085	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
111	QTUM	0.0085	Defiance Quantum ETF	Technology	LONG
4	SOXX	0.0084	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
5	SMH	0.0084	VanEck Semiconductor ETF	Technology	LONG
49	VOX	0.0084	Vanguard Communication Services ETF	Communication Services	LONG
50	FCOM	0.0084	Fidelity MSCI Communication Services In...	Communication Services	LONG
17	AOM	0.0083	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
30	XSD	0.0083	State Street SPDR S&P Semiconductor ETF	Technology	LONG
101	PSCT	0.0083	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
12	QAI	0.0082	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
27	PSI	0.0082	Invesco Semiconductors ETF	Technology	LONG
105	FTXL	0.0082	First Trust Nasdaq Semiconductor ETF	Technology	LONG
3	XLC	0.0081	State Street Communication Services Sel...	Communication Services	FLAT
110	PTF	0.0081	Invesco Dorsey Wright Technology Moment...	Technology	LONG
34	PTNQ	0.0080	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
24	PRNT	0.0078	3D Printing (The) ETF	Technology	LONG
104	RTH	0.0078	VanEck Retail ETF	Consumer Discretionary	LONG
109	IGN	0.0077			LONG
16	AOK	0.0076	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
9	ESPO	0.0074	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
33	FPXI	0.0074	First Trust International Equity Opport...	International IPO Equity	LONG
98	IYZ	0.0073	iShares U.S. Telecommunications ETF	Communication Services	FLAT
62	PTLC	0.0072	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
112	SVXY	0.0072	ProShares Short VIX Short Term Futures ...	Volatility	LONG
100	EIS	0.0070	iShares MSCI Israel ETF	Israel Equity	LONG
13	EWS	0.0065	iShares MSCI Singapore ETF	Singapore Equity	FLAT

Cluster 2 – Mixed: IJH, MDY, IVOO**Members:** 107 **Var explained:** 0.889 **Signal:** LONG since 2026-04-09 (19 days, 5.1%)**Category breakdown**

category	n_members	abs_weight
Industrials	11	0.0983
U.S. Small Cap Equity	10	0.0967
U.S. Mid Cap Equity	9	0.0876
U.S. Small Cap Value	6	0.0575
U.S. Small Cap Growth	6	0.0567
Consumer Discretionary	6	0.0512
U.S. Mid Cap Value	5	0.0484
Leveraged / Inverse	5	0.0476
U.S. Mid Cap Growth	3	0.0287
U.S. Equity	3	0.0282
U.S. Value Factor Equity	3	0.0278
U.S. Multifactor Equity	3	0.0277
Utilities	3	0.0261
U.S. Equal Weight Equity	2	0.0193
Dividend Equity	2	0.0186
U.S. Dividend Equity	2	0.0182
U.S. Small Cap Low Volatility	2	0.0178
Unknown	1	0.0098
U.S. Small Cap Fundamental Equity	1	0.0098
U.S. Mid/Small Cap Equity	1	0.0098
U.S. Small Cap ESG Equity	1	0.0098
Mid Cap Value Equities	1	0.0098
U.S. Small/Mid Cap Fundamental Equity	1	0.0097
U.S. Small Cap Multifactor Equity	1	0.0097
U.S. Mid Cap Dividend Equity	1	0.0096
U.S. ESG Equity	1	0.0096
U.S. Size Factor Equity	1	0.0095
U.S. Small/Mid Cap Equity	1	0.0095
U.S. Extended Market Equity	1	0.0095
Large Cap Blend Equities	1	0.0095
Small Cap Value Equities	1	0.0095
U.S. Mid Cap Quality Equity	1	0.0095
Value Equity	1	0.0094
Growth Equity	1	0.0094
U.S. Small Cap Dividend Equity	1	0.0094
U.S. Buyback Equity	1	0.0093
U.S. Small Cap Industrials	1	0.0092
U.S. Small Cap Momentum Equity	1	0.0092
U.S. Micro Cap Equity	1	0.0090
U.S. Moat Equity	1	0.0089
U.S. Small Cap Consumer Discretionary	1	0.0088
U.S. Mid Cap Low Volatility	1	0.0087
Real Estate	1	0.0076

Replication basket – IJH+SPSM+IWR (cor=1.00) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
IJH	0.322	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
SPSM	0.401	State Street SPDR Portfolio S&P 600 S...	U.S. Small Cap Equity	LONG
IWR	0.294	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
16	JHMM	0.0098	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
30	SCHM	0.0098	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
32	FNX	0.0098	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	LONG
35	NUSC	0.0098	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	LONG
39	MDY	0.0098	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
47	IVOO	0.0098	Vanguard S&P Mid-Cap 400 ETF		LONG
54	IJH	0.0098	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
55	SPMD	0.0098	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
57	VB	0.0098	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
69	EZM	0.0098	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	LONG
73	FNDA	0.0098	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	LONG
86	VBR	0.0098	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
105	MVV	0.0098	ProShares Ultra MidCap400	Leveraged / Inverse	LONG
21	EQAL	0.0097	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	LONG
23	IWR	0.0097	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
33	FYX	0.0097	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	LONG
36	PRFZ	0.0097	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	LONG
37	SMLF	0.0097	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
40	IJT	0.0097	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	LONG
42	VIOG	0.0097	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	LONG
45	SLYG	0.0097	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	LONG
48	VIOO	0.0097	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	LONG
51	IJR	0.0097	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
56	SPSM	0.0097	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	LONG
58	SCHA	0.0097	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
77	IWS	0.0097	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
79	IJJ	0.0097	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
81	IVOV	0.0097	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
82	MDYV	0.0097	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	LONG
98	RWK	0.0097	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	LONG
5	IWM	0.0096	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
6	RSP	0.0096	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG

7	FXR	0.0096	First Trust Industrials AlphaDEX	Industrials	FLAT
29	IMCB	0.0096	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
41	IJK	0.0096	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
44	MDYG	0.0096	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
49	VTWO	0.0096	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
67	DON	0.0096	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	LONG
76	IWN	0.0096	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
85	VTWV	0.0096	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
88	UWM	0.0096	ProShares Ultra Russell2000	Leveraged / Inverse	LONG
89	ETHO	0.0096	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	LONG
95	TNA	0.0096	Direxion Small Cap Bull 3X ETF	Leveraged / Inverse	LONG
97	JPME	0.0096	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	LONG
100	JVAL	0.0096	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
106	SDVY	0.0096	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
9	FEX	0.0095	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
25	SIZE	0.0095	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
31	VO	0.0095	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	IVOG	0.0095	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
53	VXF	0.0095	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
68	EES	0.0095	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	LONG
70	PSC	0.0095	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
80	IJS	0.0095	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
83	SLYV	0.0095	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	LONG
84	VIOV	0.0095	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	LONG
90	RWJ	0.0095	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	LONG
93	XMHQ	0.0095	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
107	ISMD	0.0095	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
13	RDVY	0.0094	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	LONG
19	CFA	0.0094	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	LONG
24	JPUS	0.0094	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	LONG
52	ISCG	0.0094	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
66	DES	0.0094	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	LONG
87	VOE	0.0094	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
96	RFV	0.0094	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	LONG
104	FYT	0.0094	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	LONG
8	PKW	0.0093	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
15	IYJ	0.0093	iShares U.S. Industrials ETF	Industrials	LONG
22	ONEV	0.0093	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
27	VIS	0.0093	Vanguard Industrials ETF	Industrials	LONG
28	FIDU	0.0093	Fidelity MSCI Industrials Index ETF	Industrials	LONG
38	IWO	0.0093	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
99	OMFS	0.0093	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
26	VLUE	0.0092	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
46	VTWG	0.0092	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
50	VBK	0.0092	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG

59	FXD	0.0092	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
91	PSCI	0.0092	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	LONG
92	PAVE	0.0092	Global X U.S. Infrastructure Developmen...	Industrials	LONG
103	XSMO	0.0092	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG
18	AIVL	0.0091	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
64	RSPD	0.0091	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
78	SMMV	0.0091	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	FLAT
101	RZV	0.0091	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	LONG
1	XLI	0.0090	State Street Industrial Select Sector S...	Industrials	LONG
12	FIW	0.0090	First Trust Water ETF	Utilities	FLAT
17	REGL	0.0090	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	FLAT
34	IWC	0.0090	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
14	MOAT	0.0089	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
20	CFO	0.0089	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	LONG
61	FTXR	0.0089	First Trust Nasdaq Transportation ETF	Industrials	LONG
10	PHO	0.0088	Invesco Water Resources ETF	Utilities	LONG
65	SMDV	0.0088	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
102	PSCD	0.0088	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
72	XTN	0.0087	State Street SPDR S&P Transportation ETF	Industrials	FLAT
74	XMLV	0.0087	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
75	XSLV	0.0087	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	LONG
94	PKB	0.0087	Invesco Building & Construction ETF	Industrials	LONG
2	XRT	0.0085	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
63	IYT	0.0085	iShares U.S. Transportation ETF	Industrials	FLAT
60	PEJ	0.0084	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
71	XHB	0.0084	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	FLAT
11	CGW	0.0082	Invesco S&P Global Water Index ETF	Utilities	FLAT
3	JETS	0.0076	U.S. Global Jets ETF	Industrials	LONG
4	REM	0.0076	iShares Mortgage Real Estate ETF	Real Estate	LONG
62	ITB	0.0076	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 3 – Mixed: VEA, SPDW, SCHF**Members:** 82 **Var explained:** 0.839 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
International Equity	21	0.2686
Developed ex-US Equity	9	0.1179

International Dividend Equity	6	0.0748
Global ex-US Equity	5	0.0656
Europe Equity	4	0.0518
Alternatives	4	0.0463
Eurozone Equity	3	0.0372
Asia-Pacific Equity	2	0.0242
Real Estate	2	0.0225
Global Equity	2	0.0222
Emerging Markets Fixed Income	2	0.0193
Currency	2	0.0181
Commodity	1	0.0132
Unknown	1	0.0130
Global ex-US Small Cap Equity	1	0.0129
Germany Equity	1	0.0124
France Equity	1	0.0124
United Kingdom Equity	1	0.0123
Italy Equity	1	0.0121
Europe Financials	1	0.0120
Sweden Equity	1	0.0120
Netherlands Equity	1	0.0120
Australia Equity	1	0.0119
Spain Equity	1	0.0116
Canada Equity	1	0.0114
Switzerland Equity	1	0.0114
Belgium Equity	1	0.0113
Austria Equity	1	0.0112
Poland Equity	1	0.0104
South Africa Equity	1	0.0099
New Zealand Equity	1	0.0091
Mexico Equity	1	0.0089

Replication basket – VEA+VGK+VYMI (cor=1.00) basket lead at lag 1: 0.004

ticker	weight_ols	name	category	etf_signal
VEA	0.479	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
VGK	0.241	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
VYMI	0.223	Vanguard International High Dividend ...	International Dividen...	FLAT

Members

ticker	weight	name	category	etf_signal
--------	--------	------	----------	------------

2	EFA	0.0133	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0133	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
63	IEFA	0.0133	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0133	Schwab International Equity ETF	Developed ex-US Equity	FLAT
65	SPDW	0.0133	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
21	DWM	0.0132	WisdomTree International Equity Fund	International Dividend E...	FLAT
34	TLTD	0.0132	FlexShares Morningstar Developed Market...	International Equity	FLAT
49	GSIE	0.0132	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
52	ESGD	0.0132	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
28	PXF	0.0131	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
38	ACWX	0.0131	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
39	CWI	0.0131	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
41	INTF	0.0131	iShares International Equity Factor ETF	International Equity	FLAT
42	IQLT	0.0131	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0131	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
58	IEUR	0.0131	iShares Core MSCI Europe ETF	Europe Equity	FLAT
59	IXUS	0.0131	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0131	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
61	VGK	0.0131	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
62	VXUS	0.0131	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0130	iShares Europe ETF	Europe Equity	FLAT
32	EFV	0.0130	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
35	JPIN	0.0130	JPMorgan Diversified Return Internation...	International Equity	FLAT
46	RODM	0.0130	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0130	Vanguard International High Dividend Yi...	International Dividend E...	FLAT
48	FNDF	0.0130	Schwab Fundamental International Equity...	International Equity	FLAT
56	SCHC	0.0130	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
75	FEU	0.0130			FLAT
24	IQDF	0.0129	FlexShares International Quality Divide...	International Equity	FLAT
31	SCZ	0.0129	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
33	FNDC	0.0129	Schwab Fundamental International Small ...	International Equity	LONG
57	VSS	0.0129	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
8	DLS	0.0128	WisdomTree International SmallCap Fund	International Equity	FLAT
19	EZU	0.0128	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
43	IVLU	0.0128	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0128	NYLI FTSE International Equity Currency...	International Equity	FLAT
29	EFG	0.0127	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
45	FEZ	0.0127	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0127	Vanguard International Dividend Appreci...	International Dividend E...	FLAT
4	FEP	0.0126	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
30	GWX	0.0126	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0126	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
1	EWG	0.0124	iShares MSCI Germany Index Fund	Germany Equity	FLAT
5	VIDI	0.0124	Vident International Equity Strategy ETF	International Equity	LONG
16	EWQ	0.0124	iShares MSCI France Index Fund	France Equity	FLAT

20	IDV	0.0124	iShares International Select Dividend E...	International Dividend E...	FLAT
53	HDEF	0.0124	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
17	EWU	0.0123	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
50	IDLV	0.0123	Invesco S&P International Developed Low...	International Equity	FLAT
79	IPKW	0.0123	Invesco International BuyBack Achievers...	International Equity	FLAT
22	EPP	0.0121	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
51	EFAV	0.0121	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
82	EWI	0.0121	iShares MSCI Italy ETF	Italy Equity	FLAT
23	EUFN	0.0120	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
71	EWD	0.0120	iShares MSCI Sweden ETF	Sweden Equity	FLAT
77	EWN	0.0120	iShares MSCI Netherlands Index Fund	Netherlands Equity	FLAT
81	BBAX	0.0120	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
13	EWA	0.0119	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0119	State Street SPDR S&P International Div...	International Dividend E...	FLAT
12	HEZU	0.0118	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
36	DBEF	0.0118	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
37	HEFA	0.0118	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
11	PID	0.0117	Invesco International Dividend Achiever...	International Dividend E...	FLAT
15	EWP	0.0116	iShares MSCI Spain ETF	Spain Equity	FLAT
9	HEDJ	0.0114	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0114	WisdomTree International Hedged Quality...	Alternatives	FLAT
14	EWL	0.0114	iShares MSCI Switzerland ETF	Switzerland Equity	FLAT
18	EWC	0.0114	iShares MSCI Canada Index Fund	Canada Equity	LONG
25	NFRA	0.0114	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
55	VNQI	0.0114	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
80	EWK	0.0113	iShares MSCI Belgium ETF	Belgium Equity	FLAT
76	EWO	0.0112	iShares MSCI Austria ETF	Austria Equity	FLAT
7	RWX	0.0111	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
26	IGF	0.0108	iShares Global Infrastructure ETF	Global Equity	LONG
70	EPOL	0.0104	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0099	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
69	FXA	0.0097	Invesco CurrencyShares Australian Dolla...	Currency	LONG
66	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
67	EMLC	0.0096	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
73	ENZL	0.0091	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWX	0.0089	iShares MSCI Mexico ETF	Mexico Equity	FLAT
78	FXC	0.0084	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG

Cluster 4 – Mixed: BND, AGG, SCHZ

Members: 59 Var explained: 0.833 Signal: FLAT

Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3225
Investment Grade Corporate Bond	13	0.2258
Aggregate Fixed Income	6	0.1100
Mortgage-Backed Fixed Income	6	0.0965
Active Fixed Income	5	0.0874
Cash / Ultra-Short Fixed Income	3	0.0462
International Fixed Income	2	0.0308
Target Maturity Bond	2	0.0304
Leveraged / Inverse Fixed Income	1	0.0178
Agency Fixed Income	1	0.0163
Taxable Municipal Bond	1	0.0162

Replication basket – BND+BLV+BIV (cor=1.00) basket lead at lag 1: 0.003

ticker	weight_ols	name	category	etf_signal
BND	0.508	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
BLV	0.132	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
BIV	0.245	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT

Members

	ticker	weight	name	category	etf_signal
26	AGG	0.0185	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0185	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
21	BIV	0.0184	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
30	SCHZ	0.0184	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0184	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
20	IUSB	0.0183	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
2	IEF	0.0182	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0181	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
7	FBND	0.0180	Fidelity Total Bond ETF	Active Fixed Income	LONG
24	VCIT	0.0180	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
43	SPTI	0.0180	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
47	VGIT	0.0180	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0179	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
46	SCHR	0.0179	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GTO	0.0178	Invesco Total Return Bond ETF	Active Fixed Income	FLAT

11	GVI	0.0178	iShares Intermediate Government/Credit ...	Fixed Income	LONG
15	SPIB	0.0178	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
40	IEI	0.0178	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
59	UST	0.0178	ProShares Ultra 7-10 Year Treasury	Leveraged / Inverse Fixe...	FLAT
27	USIG	0.0177	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
50	MBB	0.0177	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
51	VMBS	0.0177	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
5	BOND	0.0176	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
12	QLTA	0.0176	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
52	SPMB	0.0176	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0175	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
19	ISTB	0.0175	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
10	CORP	0.0174	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
8	FLCO	0.0173	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
22	BLV	0.0173	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0173	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0173	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0172	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0172	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0171	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0170	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
18	IGLB	0.0169	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0169	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
16	SPLB	0.0168	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0168	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0168	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0168	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0167	State Street DoubleLine Total Return Ta...	Active Fixed Income	LONG
1	TLT	0.0166	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0163	iShares Agency Bond ETF	Agency Fixed Income	LONG
9	BAB	0.0162	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0161	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0159	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0159	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0156	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0155	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0154	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
54	SPTS	0.0153	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
55	SCHO	0.0153	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0153	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
14	IBDR	0.0149	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0134	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0130	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0118	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 5 – Mixed: VTV, SCHV, IWD**Members:** 49 **Var explained:** 0.899 **Signal:** LONG since 2026-04-10 (18 days, 2.9%)**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	21	0.4272
U.S. Large Cap Value	10	0.2113
Value Equity	4	0.0820
U.S. Low Volatility Equity	4	0.0775
U.S. Fundamental Equity	2	0.0419
Dividend Equity	2	0.0396
U.S. Mega Cap Value	1	0.0211
U.S. Revenue Weighted Equity	1	0.0207
U.S. Large Cap Equity	1	0.0201
U.S. Quality / Factor Equity	1	0.0200
U.S. Defensive Equity	1	0.0198
Global Equity	1	0.0188

Replication basket – VTV+SDY+SPYV (cor=1.00) basket lead at lag 1: 0.010

ticker	weight_ols	name	category	etf_signal
VTV	0.466	Vanguard Value ETF	U.S. Large Cap Value	LONG
SDY	0.270	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
SPYV	0.243	State Street SPDR Portfolio S&P 500 V...	U.S. Large Cap Value	LONG

Members

	ticker	weight	name	category	etf_signal
28	VTV	0.0213	Vanguard Value ETF	U.S. Large Cap Value	LONG
16	IWD	0.0212	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
20	VONV	0.0212	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
26	SCHV	0.0212	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
10	NULV	0.0211	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
15	IWX	0.0211	iShares Russell Top 200 Value ETF	Value Equity	LONG
17	IVE	0.0211	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
18	VOOV	0.0211	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
21	MGV	0.0211	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
23	VYM	0.0211	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG

24	ILCV	0.0211	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
25	IUSV	0.0211	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
43	DTD	0.0211	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
48	DGRO	0.0211	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
5	PRF	0.0210	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
27	SPYV	0.0210	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
42	DLN	0.0210	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
14	FNDX	0.0209	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
29	FTA	0.0208	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
40	PFM	0.0208	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
6	RWL	0.0207	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
11	SDY	0.0207	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
22	SCHD	0.0207	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
2	FVD	0.0206	First Trust VL Dividend	U.S. Dividend Equity	FLAT
9	NOBL	0.0206	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
33	DVY	0.0206	iShares Select Dividend ETF	U.S. Dividend Equity	LONG
37	ONEY	0.0206	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
3	PWV	0.0205	Invesco Large Cap Value ETF	Value Equity	LONG
32	SDOG	0.0205	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
12	FDVV	0.0204	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
49	VIG	0.0204	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
38	SPYD	0.0202	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
47	LGLV	0.0202	State Street SPDR US Large Cap Low Volat...	U.S. Low Volatility Equi...	FLAT
7	DHS	0.0201	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
45	DIA	0.0201	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
39	FTCS	0.0200	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
41	OUSA	0.0200	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
8	CDC	0.0198	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	LONG
35	RPV	0.0197	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
4	FDL	0.0196	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
36	SPHD	0.0196	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	FLAT
46	USMV	0.0196	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
34	RDIV	0.0194	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
19	HDV	0.0193	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
13	LVHD	0.0192	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
30	PEY	0.0192	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	LONG
31	DIV	0.0190	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
1	ACWV	0.0188	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
44	SPLV	0.0186	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT

Cluster 6 – Mixed: IWP, VOT, IMCG**Members:** 35 **Var explained:** 0.781 **Signal:** LONG since 2026-04-10 (18 days, 10.9%)**Category breakdown**

category	n_members	abs_weight
Technology	12	0.3384
Industrials	4	0.1017
U.S. Mid Cap Growth	3	0.0944
U.S. Momentum Equity	3	0.0855
IPO Equity	2	0.0599
Convertible Bond	2	0.0598
Momentum Equity	2	0.0584
Financials	2	0.0578
Growth Equity	1	0.0304
U.S. Mid Cap Momentum Equity	1	0.0285
Consumer Discretionary	1	0.0285
U.S. Small Cap Momentum Equity	1	0.0284
Unknown	1	0.0284

Replication basket – IWP+BLOK+ARKQ (cor=0.99) basket lead at lag 1: 0.020

ticker	weight_ols	name	category	etf_signal
IWP	0.684	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
BLOK	0.089	Amplify Blockchain Technology ETF	Technology	LONG
ARKQ	0.150	ARK Autonomous Technology & Robotics ...	Technology	LONG

Members

	ticker	weight	name	category	etf_signal
22	IWP	0.0316	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
25	VOT	0.0315	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
26	IMCG	0.0314	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
18	FPX	0.0308	First Trust US Equity Opportunities ETF	IPO Equity	LONG
16	FTC	0.0306	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
20	RPG	0.0304	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
17	PDP	0.0302	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
19	CWB	0.0302	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
32	VFMO	0.0298	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG

23	ICVT	0.0296	iShares Convertible Bond ETF	Convertible Bond	LONG
7	ARKQ	0.0295	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	FINX	0.0295	Global X FinTech ETF	Financials	LONG
13	FDN	0.0293	First Trust DJ Internet Index Fund	Technology	LONG
2	SKYY	0.0292	First Trust Cloud Computing ETF	Technology	LONG
11	IPO	0.0291	Renaissance IPO ETF	IPO Equity	LONG
12	PNQI	0.0289	Invesco Nasdaq Internet ETF	Technology	LONG
6	ARKW	0.0287	ARK Next Generation Internet ETF	Technology	LONG
5	ARKK	0.0285	ARK Innovation ETF	Technology	LONG
9	IBUY	0.0285	Amplify Online Retail ETF	Consumer Discretionary	FLAT
21	XMMO	0.0285	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
24	MTUM	0.0285	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
1	IGV	0.0284	iShares Expanded Tech-Software Sector E...	Technology	LONG
14	IPAY	0.0284	Amplify Digital Payments ETF	Financials	LONG
27	DWAS	0.0284	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
35	PBS	0.0284			LONG
8	FV	0.0282	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
10	CIBR	0.0278	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
4	HACK	0.0276	Amplify Cybersecurity ETF	Technology	LONG
34	PRN	0.0275	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
31	QMOM	0.0272	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
3	BLOK	0.0264	Amplify Blockchain Technology ETF	Technology	LONG
30	XAR	0.0260	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
28	PPA	0.0249	Invesco Aerospace & Defense ETF	Industrials	FLAT
33	BLCN	0.0235	Siren NexGen Economy ETF	Technology	LONG
29	ITA	0.0234	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT

Cluster 7 – Emerging Markets Equity (19/26)**Members:** 26 **Var explained:** 0.879 **Signal:** LONG since 2026-04-10 (18 days, 10.3%)**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7408
Asia Equity	2	0.0780
Leveraged / Inverse	1	0.0407
Asia ex-Japan Equity	1	0.0401
Taiwan Equity	1	0.0342
Global Dividend Equity	1	0.0334

South Korea Equity 1 0.0328

Replication basket – IEMG+DEM+FEMS (cor=1.00) basket lead at lag 1: -0.020

ticker	weight_ols	name	category	etf_signal
IEMG	0.850	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG
DEM	0.161	WisdomTree Emerging Markets High Divi...	Emerging Markets Equi...	LONG
FEMS	0.062	First Trust Emerging Markets Small Ca...	Emerging Markets Equi...	FLAT

Members

	ticker	weight	name	category	etf_signal
21	IEMG	0.0408	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0407	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
25	EDC	0.0407	Direxion Emerging Markets Bull 3X ETF	Leveraged / Inverse	LONG
16	GEM	0.0406	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0406	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0405	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0405	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0404	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0403	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
15	EMGF	0.0402	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
5	AAXJ	0.0401	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
11	EEMA	0.0396	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0395	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0393	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0393	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0387	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0385	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0385	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0380	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0374	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0369	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0352	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0342	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0334	Global X SuperDividend ETF	Global Dividend Equity	LONG
26	FEMS	0.0332	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0328	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 8 – High Yield Bond (10) + Preferred Stock (7)**Members:** 22 **Var explained:** 0.772 **Signal:** LONG since 2026-04-09 (19 days, 1.1%)**Category breakdown**

category	n_members	abs_weight
High Yield Bond	10	0.4918
Preferred Stock	7	0.2893
Emerging Markets Fixed Income	4	0.1695
Unknown	1	0.0493

Replication basket – JNK+PFF+PCY (cor=0.99) basket lead at lag 1: 0.034

ticker	weight_ols	name	category	etf_signal
JNK	0.487	State Street SPDR Bloomberg High Yiel...	High Yield Bond	FLAT
PFF	0.257	iShares Preferred and Income Securit...	Preferred Stock	LONG
PCY	0.115	Invesco Emerging Markets Sovereign De...	Emerging Markets Fixe...	FLAT

Members

	ticker	weight	name	category	etf_signal
14	JNK	0.0502	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0501	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
20	HYLB	0.0500	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
19	BBHY	0.0499	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
17	SHYG	0.0497	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG
15	SJNK	0.0496	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
8	PHB	0.0493			FLAT
16	ANGL	0.0492	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
18	FALN	0.0492	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
6	HYS	0.0488	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
10	PFF	0.0453	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0450	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0444	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
22	VWOB	0.0443	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
12	PFXF	0.0438	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
21	PCY	0.0435	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0421	Invesco Preferred ETF	Preferred Stock	FLAT
11	PSK	0.0416	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT

4	FPE	0.0410	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0407	Invesco Financial Preferred ETF	Preferred Stock	FLAT
13	HYEM	0.0373	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
9	VRP	0.0348	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 9 – Leveraged / Inverse (16/22)**Members:** 22 **Var explained:** 0.722 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	16	0.7654
Volatility	4	0.1820
Leveraged / Inverse Fixed Income	1	0.0424
Leveraged / Inverse Commodity	1	0.0102

Replication basket – SDS+UVXY+TZA (cor=1.00) basket lead at lag 1: -0.031

ticker	weight_ols	name	category	etf_signal
SDS	0.632	ProShares UltraShort S&P500	Leveraged / Inverse	FLAT
UVXY	0.118	ProShares Ultra VIX Short Term Future...	Volatility	FLAT
TZA	0.133	Direxion Small Cap Bear 3X ETF	Leveraged / Inverse	FLAT

Members

	ticker	weight	name	category	etf_signal
16	SDS	0.0534	ProShares UltraShort S&P500	Leveraged / Inverse	FLAT
15	SH	0.0533	ProShares Short S&P500	Leveraged / Inverse	FLAT
20	SPXU	0.0533	ProShares UltraPro Short S&P500	Leveraged / Inverse	FLAT
21	SPXS	0.0533	Direxion Daily S&P 500 Bear 3X ETF	Leveraged / Inverse	FLAT
18	DXD	0.0505	ProShares UltraShort Dow30	Leveraged / Inverse	FLAT
22	SDOW	0.0505	UltraPro Short Dow30	Leveraged / Inverse	FLAT
10	QID	0.0502	ProShares UltraShort QQQ	Leveraged / Inverse	FLAT
19	PSQ	0.0502	ProShares Short QQQ	Leveraged / Inverse	FLAT
2	SQQQ	0.0501	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

11	RWM	0.0496	ProShares Short Russell2000	Leveraged / Inverse	FLAT
3	TZA	0.0495	Direxion Small Cap Bear 3X ETF	Leveraged / Inverse	FLAT
14	TWM	0.0495	ProShares UltraShort Russell2000	Leveraged / Inverse	FLAT
9	UVXY	0.0464	ProShares Ultra VIX Short Term Futures ...	Volatility	FLAT
8	VIXY	0.0462	VIX Short-Term Futures ETF	Volatility	FLAT
12	FAZ	0.0460	Direxion Financial Bear 3X ETF	Leveraged / Inverse	FLAT
5	VXZ	0.0448	iPath Series B S&P 500 VIX Mid-Term Fut...	Volatility	FLAT
7	VXX	0.0446	iPath Series B S&P 500 VIX Short-Term F...	Volatility	FLAT
17	SJB	0.0424	ProShares Short High Yield	Leveraged / Inverse Fixe...	FLAT
4	EUM	0.0403	ProShares Short MSCI Emerging Markets	Leveraged / Inverse	FLAT
6	EDZ	0.0403	Direxion Emerging Markets Bear 3X ETF	Leveraged / Inverse	FLAT
13	ERY	0.0255	Direxion Daily Energy Bear 2X ETF	Leveraged / Inverse	FLAT
1	SCO	0.0102	ProShares UltraShort Bloomberg Crude Oil	Leveraged / Inverse Comm...	FLAT

Cluster 10 – Healthcare (16/17)

Members: 17 **Var explained:** 0.764 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – FXH+XBI+VHT (cor=0.99) basket lead at lag 1: -0.011

ticker	weight_ols	name	category	etf_signal
FXH	0.364	First Trust Health Care AlphaDEX	Healthcare	FLAT
XBI	0.186	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
VHT	0.471	Vanguard Health Care ETF	Healthcare	FLAT

Members

	ticker	weight	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT

9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT
1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	FLAT
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0575	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0574	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0542	VanEck Pharmaceutical ETF	Healthcare	LONG
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0511	ARK Genomic Revolution ETF	Healthcare	LONG
11	IHF	0.0471	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 11 – Materials (13/17)

Members: 17 **Var explained:** 0.756 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Materials	13	0.7907
Consumer Staples	1	0.0590
Energy	1	0.0571
Norway Equity	1	0.0519
Industrial Metals	1	0.0413

Replication basket – GNR+IYM+PICK (cor=0.99) basket lead at lag 1: 0.043

ticker	weight_ols	name	category	etf_signal
GNR	0.410	State Street SPDR S&P Global Natural ...	Materials	FLAT
IYM	0.369	iShares U.S. Basic Materials ETF	Materials	FLAT
PICK	0.207	iShares MSCI Global Select Metals & M...	Materials	FLAT

Members

	ticker	weight	name	category	etf_signal
11	GNR	0.0653	State Street SPDR S&P Global Natural Re...	Materials	FLAT
6	MXI	0.0648	iShares Global Materials ETF	Materials	FLAT
10	GUNR	0.0643	FlexShares Global Upstream Natural Reso...	Materials	LONG
7	IYM	0.0641	iShares U.S. Basic Materials ETF	Materials	FLAT
12	PICK	0.0627	iShares MSCI Global Select Metals & Min...	Materials	FLAT
9	VAW	0.0622	Vanguard Materials ETF	Materials	FLAT
17	NANR	0.0613	State Street SPDR S&P North American Na...	Materials	FLAT
1	XLB	0.0612	State Street Materials Select Sector SP...	Materials	FLAT
8	RSPM	0.0611	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
13	SLX	0.0606	VanEck Steel ETF	Materials	LONG
4	XME	0.0605	State Street SPDR S&P Metals & Mining E...	Materials	LONG
5	MOO	0.0590	VanEck Agribusiness ETF	Consumer Staples	LONG
3	COPX	0.0587	Global X Copper Miners ETF	Materials	LONG
16	IGE	0.0571	iShares North American Natural Resource...	Energy	LONG
14	NORW	0.0519	Global X MSCI Norway ETF	Norway Equity	FLAT
2	URA	0.0438	Global X Uranium ETF	Materials	LONG
15	DBB	0.0413	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 12 – Financials (13/15)

Members: 15 **Var explained:** 0.883 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Financials	13	0.8730
Leveraged / Inverse	1	0.0694
Global Equity	1	0.0576

Replication basket – VFH+KBE+KCE (cor=1.00) basket lead at lag 1: 0.031

ticker	weight_ols	name	category	etf_signal
VFH	0.790	Vanguard Financials ETF	Financials	FLAT
KBE	0.223	State Street SPDR S&P Bank ETF	Financials	LONG
KCE	0.126	State Street SPDR S&P Capital Markets...	Financials	LONG

Members

	ticker	weight	name	category	etf_signal
10	VFH	0.0703	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0703	Fidelity MSCI Financials Index ETF	Financials	FLAT
7	IYF	0.0700	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0698	First Trust Financials AlphaDEX	Financials	LONG
8	IYG	0.0696	iShares U.S. Financial Services ETF	Financials	LONG
13	FAS	0.0694	Direxion Financial Bull 3X ETF	Leveraged / Inverse	FLAT
1	XLF	0.0691	State Street Financial Select Sector SP...	Financials	FLAT
9	KBWB	0.0681	Invesco KBW Bank ETF	Financials	LONG
3	KBE	0.0659	State Street SPDR S&P Bank ETF	Financials	LONG
15	KCE	0.0659	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0654	iShares U.S. Regional Banks ETF	Financials	LONG
14	IAI	0.0652	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0635	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0599	State Street SPDR S&P Insurance ETF	Financials	FLAT
12	PSP	0.0576	Invesco Global Listed Private Equity ETF	Global Equity	LONG

Cluster 13 – Mixed: DBP, GLTR, NUGT**Members:** 15 **Var explained:** 0.831 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Materials	5	0.3378
Gold	3	0.2011
Precious Metals	2	0.1404
Leveraged / Inverse Commodity	2	0.1338
Silver	2	0.1332
Platinum	1	0.0537

Replication basket – GLTR+GDXJ+AGQ (cor=1.00) basket lead at lag 1: 0.001

ticker	weight_ols	name	category	etf_signal
GLTR	0.384	abrdn Physical Precious Metals Basket...	Precious Metals	FLAT

GDXJ	0.423	VanEck Junior Gold Miners ETF	Materials	FLAT
AGQ	0.125	ProShares Ultra Silver	Leveraged / Inverse C...	FLAT

Members

	ticker	weight	name	category	etf_signal
12	DBP	0.0703	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0702	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
15	NUGT	0.0683	Direxion Daily Gold Miners Index Bull 2...	Leveraged / Inverse Comm...	FLAT
1	GDX	0.0682	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0682	VanEck Junior Gold Miners ETF	Materials	FLAT
13	GOAU	0.0675	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0674	Global X Silver Miners ETF	Materials	FLAT
3	GLD	0.0671	SPDR Gold Shares	Gold	FLAT
10	IAU	0.0671	iShares Gold Trust Shares	Gold	FLAT
11	SGOL	0.0669	abrdn Physical Gold Shares ETF	Gold	FLAT
4	SLV	0.0667	iShares Silver Trust	Silver	LONG
5	SILJ	0.0665	Amplify Junior Silver Miners ETF	Materials	FLAT
9	SIVR	0.0665	abrdn Physical Silver Shares ETF	Silver	LONG
14	AGQ	0.0654	ProShares Ultra Silver	Leveraged / Inverse Comm...	FLAT
8	PPLT	0.0537	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 14 – Mixed: INDA, INDY, EPI**Members:** 15 **Var explained:** 0.386 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
India Equity	3	0.3151
Unknown	2	0.1252
Latin America Equity	1	0.0848
Malaysia Equity	1	0.0765
Brazil Equity	1	0.0748
Indonesia Equity	1	0.0704
Chile Equity	1	0.0698
Philippines Equity	1	0.0660
Vietnam Equity	1	0.0486

Turkey Equity	1	0.0387
Agricultural Commodities	1	0.0251
Natural Gas	1	0.0050

Replication basket – INDA+ILF+EIDO (cor=0.97) basket lead at lag 1: -0.020

ticker	weight_ols	name	category	etf_signal
INDA	0.481	iShares MSCI India ETF	India Equity	FLAT
ILF	0.258	iShares Latin America 40 ETF	Latin America Equity	FLAT
EIDO	0.123	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT

Members

	ticker	weight	name	category	etf_signal
2	INDA	0.1058	iShares MSCI India ETF	India Equity	FLAT
4	INDY	0.1049	iShares India 50 ETF	India Equity	FLAT
5	EPI	0.1044	WisdomTree India Earnings Fund	India Equity	FLAT
14	PIN	0.1040			FLAT
7	ILF	0.0848	iShares Latin America 40 ETF	Latin America Equity	FLAT
11	EWM	0.0765	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
1	EWZ	0.0748	iShares MSCI Brazil ETF	Brazil Equity	FLAT
9	EIDO	0.0704	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
6	ECH	0.0698	iShares MSCI Chile ETF	Chile Equity	FLAT
10	EPHE	0.0660	iShares MSCI Philippines ETF	Philippines Equity	FLAT
15	VNM	0.0486	VanEck Vietnam ETF	Vietnam Equity	LONG
13	TUR	0.0387	iShares MSCI Turkey ETF	Turkey Equity	FLAT
3	DBA	0.0251	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
12	RSX	0.0212			LONG
8	UNG	0.0050	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 15 – China Equity (13/15)

Members: 15 **Var explained:** 0.804 **Signal:** LONG since 2026-04-13 (17 days, 3.7%)

Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – CXSE+CNYY+MCHI (cor=0.99) basket lead at lag 1: -0.040

ticker	weight_ols	name	category	etf_signal
CXSE	0.454	WisdomTree China ex-State-Owned Enter...	China Equity	LONG
CNYY	0.207	iShares MSCI China A ETF	China Equity	LONG
MCHI	0.320	iShares MSCI China ETF	China Equity	FLAT

Members

	ticker	weight	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	LONG
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYY	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
10	KBA	0.0594	KraneShares Bosera MSCI China A 50 Conn...	China Equity	LONG
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 16 – Energy (10/14)

Members: 14 **Var explained:** 0.861 **Signal:** LONG since 2026-04-29 (5 days, -2.9%)

Category breakdown

category	n_members	abs_weight
Energy	10	0.7397
MLP / Energy Infrastructure	4	0.2603

Replication basket – FENY+AMLP+XES (cor=1.00) basket lead at lag 1: 0.043

ticker	weight_ols	name	category	etf_signal
FENY	0.657	Fidelity MSCI Energy Index ETF	Energy	LONG
AMLP	0.200	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
XES	0.134	State Street SPDR S&P Oil & Gas Equip...	Energy	FLAT

Members

	ticker	weight	name	category	etf_signal
9	VDE	0.0758	Vanguard Energy ETF	Energy	LONG
11	FENY	0.0758	Fidelity MSCI Energy Index ETF	Energy	LONG
6	IYE	0.0756	iShares U.S. Energy ETF	Energy	LONG
1	XLE	0.0751	State Street Energy Select Sector SPDR ...	Energy	LONG
12	IEO	0.0748	iShares U.S. Oil & Gas Exploration & Pr...	Energy	LONG
10	IXC	0.0747	iShares Global Energy ETF	Energy	LONG
8	XOP	0.0739	State Street SPDR S&P Oil & Gas Explora...	Energy	LONG
14	PXE	0.0738	Invesco Energy Exploration & Production...	Energy	LONG
7	OIH	0.0702	VanEck Oil Services ETF	Energy	FLAT
13	XES	0.0699	State Street SPDR S&P Oil & Gas Equipme...	Energy	FLAT
5	MLPX	0.0688	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
3	AMLP	0.0661	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
4	MLPA	0.0655	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	EMLP	0.0600	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG

Cluster 17 – Real Estate (14/14)

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (18 days, 4.5%)

Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – SCHH+RWR+INDS (cor=1.00) basket lead at lag 1: -0.000

ticker	weight_ols	name	category	etf_signal
SCHH	0.541	Schwab U.S. REIT ETF	Real Estate	LONG
RWR	0.336	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
INDS	0.100	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Members

	ticker	weight	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 18 – Municipal Bond (13/13)

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
----------	-----------	------------

Municipal Bond	13	1
----------------	----	---

Replication basket – VTEB+HYD+CMF (cor=0.99) basket lead at lag 1: 0.188

ticker	weight_ols	name	category	etf_signal
VTEB	0.459	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
HYD	0.172	VanEck High Yield Muni ETF	Municipal Bond	LONG
CMF	0.267	iShares California Muni Bond ETF	Municipal Bond	FLAT

Members

	ticker	weight	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	FLAT
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	FLAT
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 19 – Consumer Staples (6/11)

Members: 11 **Var explained:** 0.748 **Signal:** LONG since 2026-04-27 (7 days, 0.6%)

Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5725
Utilities	4	0.3529
U.S. Small Cap Consumer Staples	1	0.0746

Replication basket – VDC+FUTY+PSCC (cor=1.00) basket lead at lag 1: -0.009

ticker	weight_ols	name	category	etf_signal
VDC	0.528	Vanguard Consumer Staples ETF	Consumer Staples	LONG
FUTY	0.361	Fidelity MSCI Utilities Index ETF	Utilities	LONG
PSCC	0.085	Invesco S&P SmallCap Consumer Staples...	U.S. Small Cap Consum...	LONG

Members

	ticker	weight	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0973	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0946	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0943	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0892	iShares U.S. Utilities ETF	Utilities	LONG
9	FUTY	0.0881	Fidelity MSCI Utilities Index ETF	Utilities	LONG
8	VPU	0.0880	Vanguard Utilities ETF	Utilities	LONG
2	XLU	0.0876	State Street Utilities Select Sector SP...	Utilities	LONG
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 20 – Broad Commodities (6/10)

Members: 10 **Var explained:** 0.912 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6001
Oil	2	0.1983
Energy Commodities	1	0.1013
Leveraged / Inverse Commodity	1	0.1003

Replication basket – GSG+UCO+DBC (cor=1.00) basket lead at lag 1: 0.055

ticker	weight_ols	name	category	etf_signal
GSG	0.398	iShares GSCI Commodity-Indexed Trust ...	Broad Commodities	FLAT
UCO	0.190	ProShares Ultra Bloomberg Crude Oil	Leveraged / Inverse C...	FLAT
DBC	0.341	Invesco DB Commodity Index Tracking F...	Broad Commodities	FLAT

Members

	ticker	weight	name	category	etf_signal
5	GSG	0.1035	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1033	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
8	COMT	0.1033	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
7	PDBC	0.1032	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
9	DBE	0.1013	Invesco DB Energy Fund	Energy Commodities	FLAT
10	UCO	0.1003	ProShares Ultra Bloomberg Crude Oil	Leveraged / Inverse Comm...	FLAT
1	USO	0.0997	United States Oil Fund	Oil	FLAT
4	DBO	0.0987	Invesco DB Oil Fund	Oil	FLAT
3	FTGC	0.0935	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
6	DJP	0.0934	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT

Cluster 21 – Inflation-Protected Fixed Income (10/10)

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (19 days, 0.6%)

Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – TIP+STPZ+SPIP (cor=1.00) basket lead at lag 1: -0.019

ticker	weight_ols	name	category	etf_signal
TIP	0.627	iShares TIPS Bond ETF	Inflation-Protected F...	LONG
STPZ	0.194	PIMCO 1-5 Year U.S. TIPS Index Exchan...	Inflation-Protected F...	LONG
SPIP	0.136	State Street SPDR Portfolio TIPS ETF	Inflation-Protected F...	LONG

Members

	ticker	weight	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDDT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	LONG
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	LONG
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG

Cluster 22 – Japan Equity (7/9)**Members:** 9 **Var explained:** 0.888 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Japan Equity	7	0.7796
Asia-Pacific Equity	2	0.2204

Replication basket – EWJ+DXJ+VPL (cor=1.00) basket lead at lag 1: -0.031

ticker	weight_ols	name	category	etf_signal
EWJ	0.409	iShares MSCI Japan Index Fund	Japan Equity	FLAT
DXJ	0.405	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
VPL	0.171	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG

Members

	ticker	weight	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	FLAT

8	BBJP	0.1140	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	FLAT
6	DBJP	0.1114	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	FLAT
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1106	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1091	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Mixed: SRLN, BKLN, FTSL

Members: 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-02 (23 days, 1.2%)

Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3633
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3123

Replication basket – SRLN+LQDH+FLOT (cor=0.97) basket lead at lag 1: -0.033

ticker	weight_ols	name	category	etf_signal
SRLN	0.346	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
LQDH	0.319	iShares Interest Rate Hedged Corporat...	Interest Rate Hedged ...	LONG
FLOT	0.322	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT

Members

	ticker	weight	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1214	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1080	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG

4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1007	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	LONG

Cluster 24 – Cash / Ultra-Short Fixed Income (9/9)**Members:** 9 **Var explained:** 0.456 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – GSY+NEAR+VNLA (cor=0.94) basket lead at lag 1: 0.029

ticker	weight_ols	name	category	etf_signal
GSY	0.376	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	FLAT
NEAR	0.124	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fi...	FLAT
VNLA	0.139	Janus Henderson Short Duration Income...	Cash / Ultra-Short Fi...	LONG

Members

	ticker	weight	name	category	etf_signal
5	GSY	0.1359	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1285	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1261	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1243	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1089	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1063	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.1031	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0881	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0787	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – Energy (5/8)**Members:** 8 **Var explained:** 0.774 **Signal:** LONG since 2026-04-10 (18 days, 12.9%)**Category breakdown**

category	n_members	abs_weight
Energy	5	0.6406
Materials	2	0.2242
Clean Energy / Climate	1	0.1352

Replication basket – QCLN+REMX+ICLN (cor=0.99) basket lead at lag 1: 0.001

ticker	weight_ols	name	category	etf_signal
QCLN	0.342	First Trust NASDAQ Clean Edge Green E...	Energy	LONG
REMX	0.205	VanEck Rare Earth and Strategic Metal...	Materials	LONG
ICLN	0.399	iShares Global Clean Energy ETF	Energy	LONG

Members

	ticker	weight	name	category	etf_signal
7	ERTH	0.1352	Invesco MSCI Sustainable Future ETF	Clean Energy / Climate	LONG
8	QCLN	0.1339	First Trust NASDAQ Clean Edge Green Ene...	Energy	LONG
6	PBW	0.1312	Invesco WilderHill Clean Energy ETF	Energy	LONG
1	ICLN	0.1308	iShares Global Clean Energy ETF	Energy	LONG
2	TAN	0.1281	Invesco Solar ETF	Energy	LONG
3	FAN	0.1165	First Trust Global Wind Energy ETF	Energy	FLAT
4	LIT	0.1153	Global X Lithium & Battery Tech ETF	Materials	LONG
5	REMX	0.1089	VanEck Rare Earth and Strategic Metals ...	Materials	LONG

Cluster 26 – Currency (5/7)**Members:** 7 **Var explained:** 0.763 **Signal:** LONG since 2026-04-13 (17 days, 0.8%)**Category breakdown**

category	n_members	abs_weight
Currency	5	0.7014
International Fixed Income	2	0.2986

Replication basket – UDN+BWX+FXF (cor=0.99) basket lead at lag 1: 0.026

ticker	weight_ols	name	category	etf_signal
UDN	0.507	Invesco DB USD Index Bearish ETF	Currency	LONG
BWX	0.337	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT
FXF	0.139	Invesco CurrencyShares Swiss Franc Tr...	Currency	FLAT

Members

	ticker	weight	name	category	etf_signal
6	UDN	0.1567	Invesco DB USD Index Bearish ETF	Currency	LONG
2	IGOV	0.1502	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1485	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1477	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1392	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1380	Invesco CurrencyShares British Pound St...	Currency	LONG
7	FXY	0.1198	Invesco CurrencyShares Japanese Yen Tru...	Currency	LONG

Cluster 27 – Mixed: UUP, EUO, TBT

Members: 6 **Var explained:** 0.459 **Signal:** LONG since 2026-05-01 (3 days, -5.2%)

Category breakdown

category	n_members	abs_weight
Leveraged / Inverse Fixed Income	2	0.3387
Leveraged / Inverse Commodity	2	0.2500
Currency	1	0.2123
Leveraged / Inverse Currency	1	0.1990

Replication basket – DUST+TBT+DZZ (cor=0.98) basket lead at lag 1: 0.006

ticker	weight_ols	name	category	etf_signal
DUST	0.194	Direxion Daily Gold Miners Index Bear...	Leveraged / Inverse C...	LONG
TBT	0.275	ProShares UltraShort Lehman 20 Year T...	Leveraged / Inverse F...	FLAT
DZZ	0.081	DB Gold Double Short ETN due February...	Leveraged / Inverse C...	FLAT

Members

	ticker	weight	name	category	etf_signal
4	UUP	0.2123	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT
6	EUO	0.1990	ProShares UltraShort Euro	Leveraged / Inverse Curr...	FLAT
5	TBT	0.1697	ProShares UltraShort Lehman 20 Year Tre...	Leveraged / Inverse Fixe...	FLAT
3	TBF	0.1690	ProShares Short 20+ Year Treasury	Leveraged / Inverse Fixe...	FLAT
2	DUST	0.1644	Direxion Daily Gold Miners Index Bear 2...	Leveraged / Inverse Comm...	LONG
1	DZZ	0.0857	DB Gold Double Short ETN due February 1...	Leveraged / Inverse Comm...	FLAT

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
RSX			cluster_14	15	LONG	0.371	0.053	0.00019	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_14	15	FLAT	0.492	0.082	0.00016	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.692	0.040	0.00006	
DZZ	DB Gold Double Short ETN due Fe...	Leveraged / Inverse C...	cluster_27	6	FLAT	0.468	0.094	0.00026	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_26	7	LONG	0.732	0.093	0.00019	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.635	0.215	0.00031	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.787	0.127	0.00026	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.825	0.022	0.00001	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_14	15	FLAT	0.591	0.178	0.00036	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.825	0.088	0.00019	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.720	0.180	0.00036	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.656	0.186	0.00043	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.793	0.158	0.00034	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	59	LONG	0.844	0.120	0.00029	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_14	15	FLAT	0.487	0.219	0.00051	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.793	0.174	0.00039	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_4	59	FLAT	0.876	0.155	0.00026	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_26	7	FLAT	0.818	0.199	0.00042	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_4	59	FLAT	0.836	0.226	0.00039	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_18	13	LONG	0.877	0.248	0.00039	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_14	15	LONG	0.452	0.334	0.00076	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_18	13	FLAT	0.877	0.261	0.00046	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_13	15	FLAT	0.750	0.269	0.00066	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_15	15	LONG	0.804	0.270	0.00065	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_11	17	FLAT	0.751	0.357	0.00076	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_15	15	LONG	0.855	0.254	0.00065	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_14	15	FLAT	0.514	0.371	0.00097	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_14	15	FLAT	0.578	0.407	0.00097	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	LONG	0.864	0.331	0.00065	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	LONG	0.920	0.223	0.00049	

Long ETF Positions

ticker	name	category	cluster	cluster_signal	entry_date	days	cum_return	prob_sell	nss
QTEC	First Trust NASDAQ-100-Technolo...	Technology	cluster_1	LONG	2026-04-08	20	24.2%	0.00	1.000
XSD	State Street SPDR S&P Semicondu...	Technology	cluster_1	LONG	2026-04-08	20	48.0%	0.00	1.000
VLUE	iShares MSCI USA Value Factor E...	U.S. Value Factor E...	cluster_2	LONG	2026-04-08	20	17.0%	0.00	1.000
SCHD	Schwab US Dividend Equity ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-17	13	2.0%	0.00	1.000
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short ...	cluster_24	FLAT	2022-10-24	885	19.6%	0.00	1.000
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short ...	cluster_24	FLAT	2022-04-20	1014	18.0%	0.00	1.000
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed...	cluster_23	LONG	2026-04-07	21	0.7%	0.00	1.000
FTXL	First Trust Nasdaq Semiconducto...	Technology	cluster_1	LONG	2026-04-07	21	49.3%	0.00	1.000
PTNQ	Pacer Trendpilot 100 ETF	Trend-Following Equ...	cluster_1	LONG	2026-04-15	15	9.1%	0.00	1.000
NXTG	First Trust Indxx NextG ETF	Technology	cluster_1	LONG	2026-04-08	20	18.5%	0.00	1.000
EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_7	LONG	2026-04-08	20	24.2%	0.00	1.000
PSI	Invesco Semiconductors ETF	Technology	cluster_1	LONG	2026-04-07	21	45.3%	0.00	1.000
ICLN	iShares Global Clean Energy ETF	Energy	cluster_25	LONG	2026-04-10	18	12.7%	0.00	1.000
NULV	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-09	19	6.2%	0.00	1.000
SOXX	iShares PHLX SOX Semiconductor ...	Technology	cluster_1	LONG	2026-04-08	20	36.8%	0.00	1.000
BSCQ	Invesco BulletShares 2026 Corpo...	Target Maturity Bond	cluster_4	FLAT	2025-04-14	266	5.0%	0.00	1.000
PSCT	Invesco S&P SmallCap Informatio...	U.S. Small Cap Tech...	cluster_1	LONG	2026-04-07	21	26.5%	0.00	1.000
QCLN	First Trust NASDAQ Clean Edge G...	Energy	cluster_25	LONG	2026-04-10	18	21.1%	0.00	1.000
SMH	VanEck Semiconductor ETF	Technology	cluster_1	LONG	2026-04-08	20	30.0%	0.00	1.000
IHF	iShares U.S. Health Care Provid...	Healthcare	cluster_10	FLAT	2026-04-09	19	8.8%	0.00	1.000
LIT	Global X Lithium & Battery Tech...	Materials	cluster_25	LONG	2026-03-31	25	23.1%	0.00	1.000
FV	First Trust Dorsey Wright Focus...	Momentum Equity	cluster_6	LONG	2026-04-13	17	8.1%	0.00	1.000
QYLD	Global X NASDAQ 100 Covered Cal...	Options Income	cluster_1	LONG	2026-04-08	20	4.2%	0.00	1.000
PBW	Invesco WilderHill Clean Energy...	Energy	cluster_25	LONG	2026-04-13	17	19.9%	0.00	1.000
TQQQ	ProShares UltraPro QQQ	Leveraged / Inverse	cluster_1	LONG	2026-04-10	18	45.6%	0.00	1.000
EWY	iShares MSCI South Korea ETF	South Korea Equity	cluster_7	LONG	2026-04-10	18	31.2%	0.00	1.000
AIQ	Global X Artificial Intelligenc...	Technology	cluster_1	LONG	2026-04-10	18	22.1%	0.00	1.000
EWM	iShares MSCI Malaysia Index Fund	Malaysia Equity	cluster_14	FLAT	2026-04-17	13	4.2%	0.00	1.000
FNDX	Schwab Fundamental U.S. Large C...	U.S. Fundamental Eq...	cluster_5	LONG	2026-04-09	19	5.6%	0.00	1.000
IYW	iShares U.S. Technology ETF	Technology	cluster_1	LONG	2026-04-09	19	19.9%	0.00	1.000
TAN	Invesco Solar ETF	Energy	cluster_25	LONG	2026-04-22	10	6.0%	0.00	1.000
QTUM	Defiance Quantum ETF	Technology	cluster_1	LONG	2026-04-09	19	23.7%	0.00	1.000
QLD	ProShares Ultra QQQ	Leveraged / Inverse	cluster_1	LONG	2026-04-10	18	28.8%	0.00	1.000
XNTK	State Street SPDR NYSE Technolo...	Technology	cluster_1	LONG	2026-04-09	19	20.5%	0.00	1.000
RDIV	Invesco S&P Ultra Dividend Reve...	U.S. Dividend Equity	cluster_5	LONG	2026-04-20	12	2.0%	0.00	1.000
STIP	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected...	cluster_21	LONG	2026-04-01	24	0.9%	0.00	1.000
QQQ	Invesco QQQ Trust, Series 1	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	13.9%	0.00	1.000
PRF	Invesco RAFI US 1000 ETF	U.S. Fundamental Eq...	cluster_5	LONG	2026-04-09	19	6.2%	0.00	1.000
XLK	State Street Technology Select ...	Technology	cluster_1	LONG	2026-04-09	19	19.7%	0.00	1.000
VTIP	Vanguard Short-Term Inflation-P...	Inflation-Protected...	cluster_21	LONG	2026-04-01	24	0.9%	0.00	1.000

FIVG			cluster_1	LONG	2026-04-02	23	22.5%	0.00	1.000
PTLC	Pacer Trendpilot US Large Cap E...	U.S. Trend-Followin...	cluster_1	LONG	2026-04-16	14	4.6%	0.00	1.000
AIA	iShares Asia 50 ETF	Asia Equity	cluster_7	LONG	2026-04-10	18	18.5%	0.00	1.000
IXN	iShares Global Tech ETF	Technology	cluster_1	LONG	2026-04-09	19	20.2%	0.00	1.000
TDIV	First Trust NASDAQ Technology D...	Technology	cluster_1	LONG	2026-04-10	18	15.9%	0.00	1.000
IWX	iShares Russell Top 200 Value E...	Value Equity	cluster_5	LONG	2026-04-09	19	5.3%	0.00	1.000
XLP	State Street Consumer Staples S...	Consumer Staples	cluster_19	LONG	2026-04-24	8	1.2%	0.00	1.000
IGM	iShares Expanded Tech Sector ETF	Technology	cluster_1	LONG	2026-04-09	19	18.7%	0.00	1.000
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_26	LONG	2026-04-30	4	0.1%	0.00	1.000
PTF	Invesco Dorsey Wright Technolog...	Technology	cluster_1	LONG	2026-04-07	21	30.7%	0.00	1.000
FXL	First Trust Technology AlphaDEX	Technology	cluster_1	LONG	2026-04-08	20	15.5%	0.00	1.000
FTEC	Fidelity MSCI Information Techn...	Technology	cluster_1	LONG	2026-04-09	19	18.5%	0.00	1.000
SKYY	First Trust Cloud Computing ETF	Technology	cluster_6	LONG	2026-04-17	13	10.6%	0.00	1.000
VGT	Vanguard Information Tech ETF	Technology	cluster_1	LONG	2026-04-09	19	18.4%	0.00	1.000
SRLN	State Street Blackstone Senior ...	Bank Loan / Floatin...	cluster_23	LONG	2026-04-02	23	1.8%	0.00	1.000
SPYD	State Street SPDR Portfolio S&P...	U.S. Dividend Equity	cluster_5	LONG	2026-04-15	15	2.8%	0.00	1.000
MTUM	iShares MSCI USA Momentum Facto...	U.S. Momentum Equity	cluster_6	LONG	2026-04-08	20	15.2%	0.00	1.000
HDV	iShares Core High Dividend ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-29	5	0.4%	0.00	1.000
RFV	Invesco S&P MidCap 400 Pure Val...	Leveraged / Inverse	cluster_2	LONG	2026-04-07	21	7.0%	0.00	1.000
XT	iShares Future Exponential Tech...	Technology	cluster_1	LONG	2026-04-10	18	10.9%	0.00	1.000
FPX	First Trust US Equity Opportuni...	IPO Equity	cluster_6	LONG	2026-04-08	20	11.7%	0.00	1.000
TDDT	FlexShares iBoxx 3 Year Target ...	Inflation-Protected...	cluster_21	LONG	2026-04-02	23	0.7%	0.00	1.000
STPZ	PIMCO 1-5 Year U.S. TIPS Index ...	Inflation-Protected...	cluster_21	LONG	2026-04-06	22	0.8%	0.00	1.000
VDC	Vanguard Consumer Staples ETF	Consumer Staples	cluster_19	LONG	2026-04-23	9	0.5%	0.00	1.000
FSTA	Fidelity MSCI Consumer Staples ...	Consumer Staples	cluster_19	LONG	2026-04-23	9	0.5%	0.00	1.000
EIS	iShares MSCI Israel ETF	Israel Equity	cluster_1	LONG	2026-04-10	18	8.7%	0.00	1.000
FXC	Invesco CurrencyShares Canadian...	Currency	cluster_3	FLAT	2026-04-17	13	0.5%	0.00	1.000
DHS	WisdomTree U.S. High Dividend F...	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	2.0%	0.00	1.000
IGN			cluster_1	LONG	2026-04-02	23	20.4%	0.00	1.000
ROBO	ROBO Global Robotics and Automa...	Technology	cluster_1	LONG	2026-04-13	17	12.6%	0.00	1.000
EEMA	iShares MSCI Emerging Markets A...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	13.7%	0.00	1.000
VOOV	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-10	18	4.3%	0.00	1.000
IVE	iShares S&P 500 Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-10	18	4.3%	0.00	1.000
BOTZ	Global X Robotics & Artificial ...	Technology	cluster_1	LONG	2026-04-14	16	9.0%	0.00	1.000
SUSA	iShares ESG Optimized MSCI USA ...	U.S. ESG Equity	cluster_1	LONG	2026-04-10	18	9.0%	0.00	1.000
SPYV	State Street SPDR Portfolio S&P...	U.S. Large Cap Value	cluster_5	LONG	2026-04-10	18	4.4%	0.00	1.000
AAXJ	iShares MSCI All Country Asia e...	Asia ex-Japan Equity	cluster_7	LONG	2026-04-10	18	13.4%	0.00	1.000
DSI	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equ...	cluster_1	LONG	2026-04-10	18	10.5%	0.00	1.000
PRN	Invesco Dorsey Wright Industria...	Industrials	cluster_6	LONG	2026-04-08	20	17.0%	0.00	1.000
ILCV	iShares Morningstar Large-Cap V...	U.S. Large Cap Value	cluster_5	LONG	2026-04-10	18	4.9%	0.00	1.000
IYK	iShares U.S. Consumer Staples E...	Consumer Staples	cluster_19	LONG	2026-04-28	6	0.4%	0.00	1.000
IWD	iShares Russell 1000 Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-09	19	5.1%	0.00	1.000
VONV	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-09	19	5.1%	0.00	1.000
RPV	Invesco S&P 500 Pure Value ETF	Value Equity	cluster_5	LONG	2026-04-09	19	2.1%	0.00	1.000
ONEQ	Fidelity Nasdaq Composite Index...	U.S. Equity	cluster_1	LONG	2026-04-09	19	13.2%	0.00	1.000

PBS			cluster_6	LONG	2026-04-13	17	12.1%	0.00	1.000
IUSV	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-10	18	4.4%	0.00	1.000
MLPA	Global X MLP ETF	MLP / Energy Infras...	cluster_16	LONG	2026-04-30	4	-2.0%	0.00	1.000
EES	WisdomTree U.S. SmallCap Fund	Small Cap Value Equ...	cluster_2	LONG	2026-04-07	21	8.6%	0.00	1.000
JVAL	JPMorgan U.S. Value Factor ETF	U.S. Value Factor E...	cluster_2	LONG	2026-04-09	19	8.1%	0.00	1.000
SPHB	Invesco S&P 500 High Beta ETF	U.S. Equity	cluster_1	LONG	2026-04-09	19	15.3%	0.00	0.999
PSCC	Invesco S&P SmallCap Consumer S...	U.S. Small Cap Cons...	cluster_19	LONG	2026-04-17	13	0.1%	0.00	0.999
FTSL	First Trust Senior Loan Fund	Bank Loan / Floatin...	cluster_23	LONG	2026-04-01	24	1.2%	0.00	0.999
ICVT	iShares Convertible Bond ETF	Convertible Bond	cluster_6	LONG	2026-04-08	20	10.3%	0.00	0.999
RWL	Invesco S&P 500 Revenue ETF	U.S. Revenue Weight...	cluster_5	LONG	2026-04-09	19	4.2%	0.00	0.999
AML	Alerian MLP ETF	MLP / Energy Infras...	cluster_16	LONG	2026-04-30	4	-2.5%	0.00	0.999
CIBR	First Trust NASDAQ Cybersecurit...	Technology	cluster_6	LONG	2026-04-17	13	5.6%	0.00	0.999
OEF	iShares S&P 100 Fund	U.S. Mega Cap Equity	cluster_1	LONG	2026-04-10	18	9.3%	0.00	0.999
ESGU	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	cluster_1	LONG	2026-04-10	18	8.4%	0.00	0.999
IOO	iShares Global 100 ETF	Global Equity	cluster_1	LONG	2026-04-09	19	9.6%	0.00	0.999
SPXL	Direxion Daily S&P 500 Bull 3X ...	Leveraged / Inverse	cluster_1	LONG	2026-04-10	18	24.9%	0.00	0.999
ONEY	State Street SPDR Russell 1000 ...	U.S. Dividend Equity	cluster_5	LONG	2026-04-09	19	2.5%	0.00	0.999
UPRO	ProShares UltraPro S&P 500	Leveraged / Inverse	cluster_1	LONG	2026-04-10	18	24.8%	0.00	0.999
QAI	NYLI Hedge Multi-Strategy Track...	Alternatives	cluster_1	LONG	2026-04-08	20	3.8%	0.00	0.999
XLG	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	cluster_1	LONG	2026-04-10	18	9.4%	0.00	0.999
QQEW	First Trust Nasdaq-100 Select E...	U.S. Equal Weight E...	cluster_1	LONG	2026-04-16	14	5.0%	0.00	0.998
CQQQ	Invesco China Technology ETF	China Equity	cluster_15	LONG	2026-04-15	15	9.4%	0.00	0.998
PWV	Invesco Large Cap Value ETF	Value Equity	cluster_5	LONG	2026-04-06	22	3.9%	0.00	0.998
PDP	Invesco Dorsey Wright Momentum ...	Momentum Equity	cluster_6	LONG	2026-04-08	20	10.5%	0.00	0.998
MGC	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	cluster_1	LONG	2026-04-10	18	8.7%	0.00	0.998
SSO	ProShares Ultra S&P500	Leveraged / Inverse	cluster_1	LONG	2026-04-10	18	16.2%	0.00	0.998
EMGF	iShares Emerging Markets Equity...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	11.9%	0.00	0.998
ILCB	iShares Morningstar Large-Cap E...	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.3%	0.00	0.998
CWB	State Street SPDR Bloomberg Con...	Convertible Bond	cluster_6	LONG	2026-04-08	20	9.7%	0.00	0.998
NULG	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-09	19	12.2%	0.00	0.998
OMFL	Invesco Russell 1000 Dynamic Mu...	U.S. Multifactor Eq...	cluster_1	LONG	2026-04-09	19	6.1%	0.00	0.998
EPS	WisdomTree U.S. LargeCap Fund	Large Cap Value Equ...	cluster_1	LONG	2026-04-09	19	7.6%	0.00	0.998
PWB	Invesco Large Cap Growth ETF	Growth Equity	cluster_1	LONG	2026-04-09	19	12.5%	0.00	0.998
SPLG			cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.998
IVV	iShares Core S&P 500 ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.998
VOO	Vanguard S&P 500 ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.998
SPY	State Street SPDR S&P 500 ETF T...	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.998
VV	Vanguard Large-Cap ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.1%	0.00	0.998
XSMO	Invesco S&P SmallCap Momentum E...	U.S. Small Cap Mome...	cluster_2	LONG	2026-04-07	21	11.8%	0.00	0.998
RWJ	Invesco S&P SmallCap 600 Revenu...	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-08	20	7.2%	0.00	0.998
IWL	iShares Russell Top 200 ETF	U.S. Equity	cluster_1	LONG	2026-04-10	18	8.4%	0.00	0.997
NURE	Nuveen Short-Term REIT ETF	Real Estate	cluster_17	LONG	2026-04-10	18	6.0%	0.00	0.997
CDC	VictoryShares US EQ Income Enha...	U.S. Defensive Equi...	cluster_5	LONG	2026-04-09	19	0.5%	0.00	0.997
IYY	iShares Dow Jones U.S. ETF	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.997
SCHB	Schwab U.S. Broad Market ETF	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.997

ITOT	iShares Core S&P Total U.S. Sto...	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	8.1%	0.00	0.997
SPTM	State Street SPDR Portfolio S&P...	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	7.9%	0.00	0.997
EDC	Direxion Emerging Markets Bull ...	Leveraged / Inverse	cluster_7	LONG	2026-04-13	17	32.8%	0.00	0.997
RPG	Invesco S&P 500 Pure Growth ETF	Growth Equity	cluster_6	LONG	2026-04-08	20	13.3%	0.00	0.997
SCHX	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.997
VTI	Vanguard Total Stock Market ETF	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.997
XSOE	WisdomTree Emerging Markets Ex-...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	11.7%	0.00	0.997
DIV	Global X Super Dividend ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-09	19	0.5%	0.00	0.996
MGV	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	cluster_5	LONG	2026-04-10	18	3.5%	0.00	0.996
IWO	iShares Russell 2000 Growth Fund	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-09	19	12.4%	0.00	0.996
VTWG	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-09	19	12.5%	0.00	0.996
KBA	KraneShares Boseri MSCI China A...	China Equity	cluster_15	LONG	2026-04-09	19	10.2%	0.00	0.996
VUG	Vanguard Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.8%	0.00	0.996
VTHR	Vanguard Russell 3000 ETF	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	7.9%	0.00	0.996
KXI	iShares Global Consumer Staples...	Consumer Staples	cluster_19	LONG	2026-04-28	6	1.8%	0.00	0.996
IWB	iShares Russell 1000 ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	7.8%	0.00	0.996
SDOG	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-08	20	1.6%	0.00	0.996
VFMO	Vanguard U.S. Momentum Factor E...	U.S. Momentum Equity	cluster_6	LONG	2026-04-08	20	10.5%	0.00	0.995
TNA	Direxion Small Cap Bull 3X ETF	Leveraged / Inverse	cluster_2	LONG	2026-04-09	19	29.5%	0.00	0.995
GMF	State Street SPDR S&P Emerging ...	Asia Equity	cluster_7	LONG	2026-04-13	17	8.1%	0.00	0.995
QDF	FlexShares Quality Dividend Ind...	Dividend Equity	cluster_1	LONG	2026-04-10	18	6.7%	0.00	0.995
CATH	Global X S&P 500 Catholic Value...	Global Equity	cluster_1	LONG	2026-04-10	18	8.0%	0.00	0.995
MGK	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	cluster_1	LONG	2026-04-10	18	12.1%	0.00	0.995
EEM	iShares MSCI Emerging Index Fund	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	11.4%	0.00	0.995
SPYX	State Street SPDR S&P 500 Fossi...	U.S. Equity	cluster_1	LONG	2026-04-10	18	8.2%	0.00	0.995
TIPX	State Street SPDR Bloomberg 1a?...	Inflation-Protected...	cluster_21	LONG	2026-04-09	19	0.7%	0.01	0.995
IWV	iShares Russell 3000 Fund	U.S. Total Market E...	cluster_1	LONG	2026-04-10	18	7.8%	0.01	0.995
HACK	Amplify Cybersecurity ETF	Technology	cluster_6	LONG	2026-04-20	12	5.2%	0.00	0.995
VONE	Vanguard Russell 1000 ETF	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-10	18	7.7%	0.01	0.995
TILT	FlexShares Mornigstar US Market...	U.S. Multifactor Eq...	cluster_1	LONG	2026-04-09	19	6.8%	0.01	0.995
IEMG	iShares Core MSCI Emerging Mark...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	11.2%	0.01	0.995
SCHG	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	10.8%	0.01	0.994
PSC	Principal U.S. Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-09	19	7.7%	0.01	0.994
SCHV	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-09	19	3.7%	0.01	0.994
SCHA	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-08	20	9.6%	0.01	0.994
IMCG	iShares Morningstar Mid-Cap Gro...	U.S. Mid Cap Growth	cluster_6	LONG	2026-04-09	19	9.2%	0.01	0.994
VIOG	Vanguard S&P Small-Cap 600 Grow...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	20	7.6%	0.01	0.994
IVW	iShares S&P 500 Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.3%	0.01	0.994
ILCG	iShares Morningstar Large-Cap G...	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.9%	0.01	0.994
UWM	ProShares Ultra Russell2000	Leveraged / Inverse	cluster_2	LONG	2026-04-09	19	19.2%	0.01	0.994
SPYG	State Street SPDR Portfolio S&P...	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.3%	0.01	0.994
VOOG	Vanguard S&P 500 Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.3%	0.01	0.994
IWM	iShares Russell 2000 Index Fund	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-09	19	9.5%	0.01	0.993
VTWO	Vanguard Russell 2000 ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-09	19	9.5%	0.01	0.993
ESGE	iShares ESG Aware MSCI EM ETF	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	11.1%	0.01	0.993

IUSG	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-10	18	11.1%	0.01	0.993
REZ	iShares Residential and Multise...	Real Estate	cluster_17	LONG	2026-04-14	16	3.6%	0.01	0.993
JHML	John Hancock Multifactor Large ...	U.S. Large Cap Equi...	cluster_1	LONG	2026-04-09	19	6.5%	0.01	0.993
SLYG	State Street SPDR S&P 600 Small...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	20	7.6%	0.01	0.993
VTV	Vanguard Value ETF	U.S. Large Cap Value	cluster_5	LONG	2026-04-09	19	2.5%	0.01	0.992
PSCH	Invesco S&P SmallCap Health Car...	U.S. Small Cap Heal...	cluster_10	FLAT	2026-04-08	20	7.4%	0.01	0.992
CNYA	iShares MSCI China A ETF	China Equity	cluster_15	LONG	2026-04-13	17	5.9%	0.01	0.992
SPSM	State Street SPDR Portfolio S&P...	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-08	20	7.1%	0.01	0.992
DVY	iShares Select Dividend ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-08	20	0.5%	0.01	0.992
IJT	iShares S&P SmallCap 600 Growth...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	20	7.6%	0.01	0.992
VIOO	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-08	20	7.0%	0.01	0.992
IJR	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-08	20	7.0%	0.01	0.991
ASHR	Xtrackers Harvest CSI 300 China...	China Equity	cluster_15	LONG	2026-04-13	17	5.4%	0.01	0.991
MLPX	Global X MLP & Energy Infrastru...	MLP / Energy Infras...	cluster_16	LONG	2026-04-30	4	-3.0%	0.00	0.991
EMLP	First Trust North American Ener...	MLP / Energy Infras...	cluster_16	LONG	2026-04-29	5	-0.5%	0.00	0.990
VOT	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	cluster_6	LONG	2026-04-13	17	7.4%	0.01	0.990
SCHM	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	7.7%	0.01	0.990
ARKW	ARK Next Generation Internet ETF	Technology	cluster_6	LONG	2026-04-14	16	11.5%	0.01	0.990
BLCN	Siren NexGen Economy ETF	Technology	cluster_6	LONG	2026-04-14	16	11.0%	0.01	0.990
PAVE	Global X U.S. Infrastructure De...	Industrials	cluster_2	LONG	2026-04-09	19	6.9%	0.01	0.989
REMX	VanEck Rare Earth and Strategic...	Materials	cluster_25	LONG	2026-04-10	18	15.4%	0.01	0.989
IMCB	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	5.6%	0.01	0.989
QUS	State Street SPDR MSCI USA Stra...	U.S. Quality / Fact...	cluster_1	LONG	2026-04-13	17	3.1%	0.01	0.988
IPO	Renaissance IPO ETF	IPO Equity	cluster_6	LONG	2026-04-10	18	18.4%	0.01	0.988
TDTF	FlexShares iBoxx 5 Year Target ...	Inflation-Protected...	cluster_21	LONG	2026-04-10	18	0.8%	0.01	0.988
PFXF	VanEck Preferred Securities ex ...	Preferred Stock	cluster_8	LONG	2026-04-13	17	4.2%	0.01	0.988
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_15	LONG	2026-04-10	18	12.8%	0.01	0.987
PEY	Invesco High Yield Equity Divid...	U.S. Dividend Equity	cluster_5	LONG	2026-04-06	22	3.0%	0.01	0.987
GSLC	Goldman Sachs ActiveBeta U.S. L...	Commodity	cluster_1	LONG	2026-04-10	18	7.2%	0.01	0.987
EEMV	iShares MSCI Emerging Markets M...	Emerging Markets Eq...	cluster_7	LONG	2026-04-09	19	7.8%	0.01	0.987
VO	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-10	18	5.4%	0.01	0.986
IWC	iShares Microcap ETF	U.S. Micro Cap Equi...	cluster_2	LONG	2026-04-08	20	11.9%	0.01	0.986
GEM	Goldman Sachs ActiveBeta Emergi...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	10.5%	0.01	0.986
VYM	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-09	19	2.8%	0.01	0.985
IGV	iShares Expanded Tech-Software ...	Technology	cluster_6	LONG	2026-04-20	12	1.8%	0.01	0.985
SLX	VanEck Steel ETF	Materials	cluster_11	FLAT	2026-04-08	20	10.9%	0.01	0.985
PRFZ	Invesco RAFI US 1500 Small-Mid ...	U.S. Small/Mid Cap ...	cluster_2	LONG	2026-04-09	19	7.3%	0.02	0.985
DLN	WisdomTree U.S. LargeCap Divide...	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	3.6%	0.02	0.985
SLYV	State Street SPDR S&P 600 Small...	U.S. Small Cap Value	cluster_2	LONG	2026-04-08	20	6.3%	0.02	0.985
FYX	First Trust Small Cap Core Alph...	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-07	21	10.6%	0.02	0.984
PRNT	3D Printing (The) ETF	Technology	cluster_1	LONG	2026-04-14	16	8.0%	0.02	0.984
VIOV	Vanguard S&P Small-Cap 600 Valu...	U.S. Small Cap Value	cluster_2	LONG	2026-04-08	20	6.4%	0.02	0.984
IJS	iShares S&P SmallCap 600 Value ...	U.S. Small Cap Value	cluster_2	LONG	2026-04-08	20	6.4%	0.02	0.984
LRGF	iShares U.S. Equity Factor ETF	U.S. Multifactor Eq...	cluster_1	LONG	2026-04-09	19	7.1%	0.02	0.984
BLOK	Amplify Blockchain Technology E...	Technology	cluster_6	LONG	2026-04-10	18	22.1%	0.02	0.983

DTD	WisdomTree U.S. Total Dividend ...	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	3.5%	0.02	0.982
FTC	First Trust Large Cap Growth Al...	Technology	cluster_6	LONG	2026-04-09	19	9.0%	0.02	0.982
ISMD	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap ...	cluster_2	LONG	2026-04-08	20	8.1%	0.02	0.982
IWY	iShares Russell Top 200 Growth ...	Growth Equity	cluster_1	LONG	2026-04-13	17	8.5%	0.02	0.979
FDN	First Trust DJ Internet Index F...	Technology	cluster_6	LONG	2026-04-08	20	8.1%	0.02	0.978
VBK	Vanguard Small-Cap Growth ETF	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-09	19	9.9%	0.02	0.978
ACWI	iShares MSCI ACWI ETF	Global Equity	cluster_1	LONG	2026-04-10	18	7.1%	0.02	0.978
IWF	iShares Russell 1000 Growth Fund	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-13	17	8.3%	0.02	0.976
DWAS	Invesco Dorsey Wright SmallCap ...	U.S. Small Cap Mome...	cluster_6	LONG	2026-04-07	21	14.3%	0.02	0.976
VONG	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Grow...	cluster_1	LONG	2026-04-13	17	8.3%	0.02	0.976
SPHQ	Invesco S&P 500 Quality ETF	U.S. Quality / Fact...	cluster_1	LONG	2026-04-10	18	5.0%	0.02	0.976
SPGM	State Street SPDR Portfolio MSC...	Global Equity	cluster_1	LONG	2026-04-09	19	7.0%	0.02	0.975
IWN	iShares Russell 2000 Value ETF	U.S. Small Cap Value	cluster_2	LONG	2026-04-08	20	7.2%	0.03	0.975
VPL	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	cluster_22	FLAT	2026-04-09	19	8.2%	0.03	0.973
CRBN	iShares Low Carbon Optimized MS...	Global Equity	cluster_1	LONG	2026-04-10	18	7.6%	0.03	0.973
PSR	Invesco Active U.S. Real Estate...	Real Estate	cluster_17	LONG	2026-04-10	18	5.0%	0.03	0.972
XMMO	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Moment...	cluster_6	LONG	2026-04-07	21	14.0%	0.03	0.972
NFRA	FlexShares STOXX Global Broad I...	Global Equity	cluster_3	FLAT	2026-04-10	18	1.8%	0.03	0.972
VTWV	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	cluster_2	LONG	2026-04-08	20	7.2%	0.03	0.972
VXF	Vanguard Extended Market ETF	U.S. Extended Marke...	cluster_2	LONG	2026-04-09	19	8.1%	0.03	0.971
VT	Vanguard Total World Stock Inde...	Global Equity	cluster_1	LONG	2026-04-10	18	7.0%	0.03	0.970
URTH	iShares MSCI World ETF	Global Equity	cluster_1	LONG	2026-04-10	18	6.5%	0.03	0.970
DEM	WisdomTree Emerging Markets Hig...	Emerging Markets Eq...	cluster_7	LONG	2026-04-08	20	6.4%	0.03	0.970
FEX	First Trust Large Cap Core Alph...	Large Cap Blend Equ...	cluster_2	LONG	2026-04-09	19	4.6%	0.03	0.966
FEM	First Trust Emerging Markets Al...	Emerging Markets Eq...	cluster_7	LONG	2026-04-07	21	11.4%	0.04	0.964
FNDA	Schwab Fundamental U.S. Small C...	U.S. Small Cap Fund...	cluster_2	LONG	2026-04-09	19	6.1%	0.04	0.962
PFM	Invesco Dividend Achievers ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	3.5%	0.04	0.959
ICF	iShares Select U.S. REIT ETF	Real Estate	cluster_17	LONG	2026-04-09	19	4.9%	0.04	0.958
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short ...	cluster_24	FLAT	2026-04-06	22	0.5%	0.04	0.956
AOA	iShares Core 80/20 Aggressive A...	Asset Allocation	cluster_1	LONG	2026-04-10	18	5.5%	0.04	0.955
OMFS	Invesco Russell 2000 Dynamic Mu...	U.S. Equity	cluster_2	LONG	2026-04-08	20	7.7%	0.05	0.954
VOE	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	cluster_2	LONG	2026-04-09	19	2.0%	0.05	0.953
SIZE	iShares MSCI USA Size Factor ETF	U.S. Size Factor Eq...	cluster_2	LONG	2026-04-10	18	5.3%	0.05	0.953
LVHD	Franklin U.S. Low Volatility Hi...	U.S. Low Volatility...	cluster_5	LONG	2026-04-28	6	-0.1%	0.05	0.952
VIG	Vanguard Div Appreciation ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	3.8%	0.05	0.952
QMOM	Alpha Architect U.S. Quantitati...	U.S. Momentum Equity	cluster_6	LONG	2026-04-08	20	9.7%	0.05	0.951
IWS	iShares Russell Mid-Cap Value E...	U.S. Mid Cap Value	cluster_2	LONG	2026-04-08	20	4.8%	0.05	0.950
EWX	State Street SPDR S&P Emerging ...	Emerging Markets Eq...	cluster_7	LONG	2026-04-09	19	8.2%	0.05	0.949
PKB	Invesco Building & Construction...	Industrials	cluster_2	LONG	2026-04-10	18	5.8%	0.05	0.947
RZV	Invesco S&P SmallCap 600 Pure V...	Leveraged / Inverse	cluster_2	LONG	2026-04-09	19	6.2%	0.05	0.947
IWR	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	4.9%	0.06	0.945
USRT	iShares Core U.S. REIT ETF	Real Estate	cluster_17	LONG	2026-04-10	18	5.0%	0.06	0.944
SPEM	State Street SPDR Portfolio Eme...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	7.1%	0.06	0.941
ISCG	iShares Morningstar Small-Cap G...	Growth Equity	cluster_2	LONG	2026-04-09	19	7.6%	0.06	0.940
VB	Vanguard Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-09	19	6.5%	0.06	0.940

SCHP	Schwab U.S. TIPS ETF	Inflation-Protected...	cluster_21	LONG	2026-04-13	17	0.4%	0.06	0.938
FNX	First Trust Mid Cap Core AlphaD...	U.S. Mid/Small Cap ...	cluster_2	LONG	2026-04-09	19	4.9%	0.06	0.936
TIP	iShares TIPS Bond ETF	Inflation-Protected...	cluster_21	LONG	2026-04-13	17	0.4%	0.06	0.936
IGHG	ProShares Investment Grade-Inte...	Interest Rate Hedge...	cluster_23	LONG	2026-04-06	22	1.2%	0.06	0.935
SMLF	iShares U.S. Small-Cap Equity F...	U.S. Small Cap Mult...	cluster_2	LONG	2026-04-08	20	7.9%	0.07	0.935
AOR	iShares Core 60/40 Balanced All...	Asset Allocation	cluster_1	LONG	2026-04-10	18	4.3%	0.07	0.934
PSCI	Invesco S&P SmallCap Industrial...	U.S. Small Cap Indu...	cluster_2	LONG	2026-04-09	19	5.5%	0.07	0.932
PFF	iShares Preferred and Income Se...	Preferred Stock	cluster_8	LONG	2026-04-13	17	2.5%	0.07	0.931
VWO	Vanguard FTSE Emerging Markets ...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	7.0%	0.07	0.931
FXB	Invesco CurrencyShares British ...	Currency	cluster_26	LONG	2026-04-10	18	1.0%	0.07	0.928
DUST	Direxion Daily Gold Miners Inde...	Leveraged / Inverse...	cluster_27	LONG	2026-04-30	4	-10.1%	0.00	0.928
PXH	Invesco RAFI Emerging Markets E...	Emerging Markets Eq...	cluster_7	LONG	2026-04-09	19	6.2%	0.08	0.922
SCHH	Schwab U.S. REIT ETF	Real Estate	cluster_17	LONG	2026-04-10	18	4.4%	0.08	0.921
DGRO	iShares Core Dividend Growth ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	2.5%	0.08	0.921
FXA	Invesco CurrencyShares Australi...	Currency	cluster_3	FLAT	2026-04-13	17	2.0%	0.08	0.921
SCHE	Schwab Emerging Markets Equity ...	Emerging Markets Eq...	cluster_7	LONG	2026-04-10	18	6.8%	0.09	0.914
ETHO	Amplify Etho Climate Leadership...	U.S. ESG Equity	cluster_2	LONG	2026-04-08	20	6.8%	0.09	0.909
DGRW	WisdomTree U.S. Quality Dividen...	U.S. Dividend Equity	cluster_1	LONG	2026-04-13	17	4.5%	0.09	0.909
OUSA	ALPS O'Shares U.S. Quality Divi...	Dividend Equity	cluster_5	LONG	2026-04-15	15	1.2%	0.09	0.909
QUAL	iShares MSCI USA Quality Factor...	U.S. Quality / Fact...	cluster_1	LONG	2026-04-13	17	4.4%	0.09	0.906
RWR	State Street SPDR Dow Jones REI...	Real Estate	cluster_17	LONG	2026-04-10	18	4.6%	0.10	0.905
JPME	JPMorgan Diversified Return U.S...	U.S. Equity	cluster_2	LONG	2026-04-08	20	3.3%	0.10	0.904
XLRE	State Street Real Estate Select...	Real Estate	cluster_17	LONG	2026-04-10	18	4.5%	0.10	0.903
RSP	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight E...	cluster_2	LONG	2026-04-13	17	3.0%	0.10	0.899
GWX	State Street SPDR S&P Internati...	International Equity	cluster_3	FLAT	2026-04-10	18	5.6%	0.10	0.896
IJK	iShares S&P Mid-Cap 400 Growth ...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	19	7.5%	0.10	0.895
IWP	iShares Russell Midcap Growth E...	U.S. Mid Cap Growth	cluster_6	LONG	2026-04-14	16	3.3%	0.11	0.894
MDYG	State Street SPDR S&P 400 Mid C...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	19	7.5%	0.11	0.893
ENZL	iShares MSCI New Zealand ETF	New Zealand Equity	cluster_3	FLAT	2026-04-30	4	3.1%	0.11	0.892
DES	WisdomTree U.S. SmallCap Divide...	U.S. Small Cap Divi...	cluster_2	LONG	2026-04-07	21	6.6%	0.11	0.892
SDY	State Street SPDR S&P Dividend ...	U.S. Dividend Equity	cluster_5	LONG	2026-04-20	12	0.0%	0.11	0.892
FNDE	Schwab Fundamental Emerging Mar...	Emerging Markets Eq...	cluster_7	LONG	2026-04-09	19	5.7%	0.11	0.889
VIDI	Vident International Equity Str...	International Equity	cluster_3	FLAT	2026-04-08	20	4.9%	0.11	0.889
ARKQ	ARK Autonomous Technology & Rob...	Technology	cluster_6	LONG	2026-04-13	17	10.2%	0.11	0.887
XME	State Street SPDR S&P Metals & ...	Materials	cluster_11	FLAT	2026-04-10	18	9.3%	0.11	0.886
SPIP	State Street SPDR Portfolio TIP...	Inflation-Protected...	cluster_21	LONG	2026-04-13	17	0.4%	0.12	0.881
SPMD	State Street SPDR Portfolio S&P...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	5.6%	0.12	0.880
IVOG	Vanguard S&P Mid-Cap 400 Growth...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	19	7.4%	0.12	0.879
SVXY	ProShares Short VIX Short Term ...	Volatility	cluster_1	LONG	2026-04-13	17	2.7%	0.12	0.878
IJH	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	5.6%	0.12	0.878
DIA	State Street SPDR Dow Jones Ind...	U.S. Large Cap Equi...	cluster_5	LONG	2026-04-10	18	4.2%	0.12	0.876
IVOO	Vanguard S&P Mid-Cap 400 ETF		cluster_2	LONG	2026-04-09	19	5.6%	0.12	0.876
MDY	State Street SPDR S&P MIDCAP 40...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	5.5%	0.12	0.875
JHMM	John Hancock Multifactor Mid Ca...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	19	4.6%	0.13	0.869
VNQ	Vanguard Real Estate ETF	Real Estate	cluster_17	LONG	2026-04-10	18	4.4%	0.13	0.866

AOM	iShares Core 40/60 Moderate All...	Asset Allocation	cluster_1	LONG	2026-04-10	18	2.9%	0.13	0.865
IYR	iShares U.S. Real Estate ETF	Real Estate	cluster_17	LONG	2026-04-10	18	4.3%	0.14	0.864
FREL	Fidelity MSCI Real Estate Index...	Real Estate	cluster_17	LONG	2026-04-10	18	4.4%	0.14	0.862
AIVL	WisdomTree U.S. AI Enhanced Val...	U.S. Value Factor E...	cluster_2	LONG	2026-04-09	19	2.2%	0.14	0.861
USMV	iShares MSCI USA Min Vol Factor...	U.S. Low Volatility...	cluster_5	LONG	2026-04-20	12	-0.8%	0.14	0.858
MVV	ProShares Ultra MidCap400	Leveraged / Inverse	cluster_2	LONG	2026-04-09	19	10.8%	0.14	0.857
URA	Global X Uranium ETF	Materials	cluster_11	FLAT	2026-04-13	17	11.7%	0.15	0.853
NUSC	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG ...	cluster_2	LONG	2026-04-09	19	5.2%	0.15	0.850
EZM	WisdomTree U.S. MidCap Fund	Mid Cap Value Equit...	cluster_2	LONG	2026-04-09	19	4.4%	0.15	0.849
REET	iShares Global REIT ETF	Real Estate	cluster_17	LONG	2026-04-10	18	4.4%	0.16	0.844
HYEM	VanEck Emerging Markets High Yi...	Emerging Markets Fi...	cluster_8	LONG	2026-04-13	17	1.2%	0.16	0.836
FDIV	Fidelity High Dividend ETF	U.S. Dividend Equity	cluster_5	LONG	2026-04-10	18	4.2%	0.17	0.834
ARKK	ARK Innovation ETF	Technology	cluster_6	LONG	2026-04-14	16	6.3%	0.17	0.825
IJJ	iShares S&P Mid-Cap 400 Value E...	U.S. Mid Cap Value	cluster_2	LONG	2026-04-09	19	3.6%	0.18	0.821
MDYV	State Street SPDR S&P 400 Mid C...	U.S. Mid Cap Value	cluster_2	LONG	2026-04-09	19	3.6%	0.18	0.819
FYT	First Trust Small Cap Value Alp...	Value Equity	cluster_2	LONG	2026-04-07	21	6.9%	0.18	0.817
VBR	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	cluster_2	LONG	2026-04-09	19	4.0%	0.18	0.815
IVOV	Vanguard S&P Mid-Cap 400 Value ...	U.S. Mid Cap Value	cluster_2	LONG	2026-04-09	19	3.7%	0.19	0.813
VCR	Vanguard Consumer Discretion ETF	Consumer Discretion...	cluster_1	LONG	2026-04-13	17	5.6%	0.19	0.809
ARKG	ARK Genomic Revolution ETF	Healthcare	cluster_10	FLAT	2026-04-14	16	6.0%	0.19	0.809
FDIS	Fidelity MSCI Consumer Discreti...	Consumer Discretion...	cluster_1	LONG	2026-04-13	17	5.6%	0.19	0.807
AOK	iShares Core 30/70 Conservative...	Asset Allocation	cluster_1	LONG	2026-04-10	18	2.3%	0.19	0.806
FPXI	First Trust International Equit...	International IPO E...	cluster_1	LONG	2026-04-09	19	11.0%	0.19	0.805
VEU	Vanguard FTSE All World Ex US E...	Global ex-US Equity	cluster_3	FLAT	2026-04-09	19	5.6%	0.20	0.800
JPUS	JPMorgan Diversified Return U.S...	U.S. Multifactor Eq...	cluster_2	LONG	2026-04-08	20	2.1%	0.20	0.797
VXUS	Vanguard Total International St...	Global ex-US Equity	cluster_3	FLAT	2026-04-09	19	5.6%	0.21	0.793
XLY	State Street Consumer Discretio...	Consumer Discretion...	cluster_1	LONG	2026-04-14	16	2.9%	0.21	0.788
PNQI	Invesco Nasdaq Internet ETF	Technology	cluster_6	LONG	2026-04-14	16	3.7%	0.21	0.788
VIS	Vanguard Industrials ETF	Industrials	cluster_2	LONG	2026-04-10	18	4.1%	0.21	0.786
FDL	First Trust Morningstar ETF	U.S. Dividend Equity	cluster_5	LONG	2026-05-01	3	-1.3%	0.10	0.785
FIDU	Fidelity MSCI Industrials Index...	Industrials	cluster_2	LONG	2026-04-10	18	4.0%	0.22	0.785
PXF	Invesco RAFI Developed Markets ...	International Equity	cluster_3	FLAT	2026-04-08	20	3.8%	0.22	0.781
RTH	VanEck Retail ETF	Consumer Discretion...	cluster_1	LONG	2026-04-09	19	1.6%	0.22	0.781
CFA	VictoryShares US 500 Volatility...	U.S. Multifactor Eq...	cluster_2	LONG	2026-04-10	18	2.8%	0.23	0.774
FINX	Global X FinTech ETF	Financials	cluster_6	LONG	2026-04-15	15	1.0%	0.23	0.771
DON	WisdomTree U.S. MidCap Dividend...	U.S. Mid Cap Divide...	cluster_2	LONG	2026-04-09	19	2.1%	0.23	0.770
IXUS	iShares Core MSCI Total Interna...	Global ex-US Equity	cluster_3	FLAT	2026-04-09	19	5.6%	0.23	0.768
UDN	Invesco DB USD Index Bearish ETF	Currency	cluster_26	LONG	2026-04-10	18	0.8%	0.24	0.764
VSS	Vanguard FTSE All-Wld ex-US SmC...	Global ex-US Small ...	cluster_3	FLAT	2026-04-10	18	5.2%	0.24	0.762
CFO	VictoryShares US 500 Enhanced V...	U.S. Multifactor Eq...	cluster_2	LONG	2026-04-10	18	2.9%	0.24	0.759
CXSE	WisdomTree China ex-State-Owned...	China Equity	cluster_15	LONG	2026-04-13	17	4.3%	0.24	0.759
RWO	State Street SPDR Dow Jones Glo...	Real Estate	cluster_17	LONG	2026-04-13	17	3.5%	0.24	0.757
EQAL	Invesco Russell 1000 Equal Weig...	U.S. Equal Weight E...	cluster_2	LONG	2026-04-08	20	2.6%	0.25	0.753
CWI	State Street SPDR MSCI ACWI ex-...	Global ex-US Equity	cluster_3	FLAT	2026-04-09	19	5.3%	0.25	0.749
ACWX	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	cluster_3	FLAT	2026-04-09	19	5.4%	0.26	0.740

VOX	Vanguard Communication Services...	Communication Servi...	cluster_1	LONG	2026-04-10	18	5.2%	0.26	0.740
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	FLAT	2026-04-09	19	0.2%	0.27	0.732
IGBH	iShares Interest Rate Hedged Lo...	Interest Rate Hedge...	cluster_23	LONG	2026-04-06	22	1.7%	0.27	0.727
FCOM	Fidelity MSCI Communication Ser...	Communication Servi...	cluster_1	LONG	2026-04-10	18	5.1%	0.28	0.721
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_14	FLAT	2026-04-09	19	6.5%	0.28	0.719
FPE	First Trust Preferred Securitie...	Preferred Stock	cluster_8	LONG	2026-04-13	17	0.8%	0.29	0.714
XLI	State Street Industrial Select ...	Industrials	cluster_2	LONG	2026-04-10	18	3.1%	0.30	0.705
PSP	Invesco Global Listed Private E...	Global Equity	cluster_12	FLAT	2026-04-13	17	5.4%	0.30	0.695
IGF	iShares Global Infrastructure E...	Global Equity	cluster_3	FLAT	2026-04-24	8	0.3%	0.29	0.692
EWC	iShares MSCI Canada Index Fund	Canada Equity	cluster_3	FLAT	2026-04-09	19	3.1%	0.31	0.688
IPAY	Amplify Digital Payments ETF	Financials	cluster_6	LONG	2026-04-14	16	2.3%	0.32	0.678
PXE	Invesco Energy Exploration & Pr...	Energy	cluster_16	LONG	2026-04-30	4	-4.8%	0.02	0.674
SDIV	Global X SuperDividend ETF	Global Dividend Equ...	cluster_7	LONG	2026-04-07	21	4.3%	0.33	0.673
LQDH	iShares Interest Rate Hedged Co...	Interest Rate Hedge...	cluster_23	LONG	2026-04-02	23	1.2%	0.33	0.672
FNDC	Schwab Fundamental Internationa...	International Equity	cluster_3	FLAT	2026-04-09	19	3.7%	0.33	0.669
KCE	State Street SPDR S&P Capital M...	Financials	cluster_12	FLAT	2026-04-09	19	6.6%	0.34	0.660
XOP	State Street SPDR S&P Oil & Gas...	Energy	cluster_16	LONG	2026-04-30	4	-5.0%	0.01	0.659
RDVY	First Trust Rising Dividend Ach...	U.S. Dividend Equity	cluster_2	LONG	2026-04-09	19	4.0%	0.34	0.656
VDE	Vanguard Energy ETF	Energy	cluster_16	LONG	2026-05-01	3	-3.0%	0.02	0.639
ERTH	Invesco MSCI Sustainable Future...	Clean Energy / Clim...	cluster_25	LONG	2026-04-09	19	4.7%	0.37	0.632
IEO	iShares U.S. Oil & Gas Explorat...	Energy	cluster_16	LONG	2026-04-30	4	-5.1%	0.01	0.625
FENY	Fidelity MSCI Energy Index ETF	Energy	cluster_16	LONG	2026-05-01	3	-3.0%	0.02	0.623
FUTY	Fidelity MSCI Utilities Index E...	Utilities	cluster_19	LONG	2026-04-29	5	0.2%	0.33	0.622
VPU	Vanguard Utilities ETF	Utilities	cluster_19	LONG	2026-04-29	5	0.2%	0.33	0.621
XLU	State Street Utilities Select S...	Utilities	cluster_19	LONG	2026-04-29	5	0.1%	0.31	0.619
RWK	Invesco S&P MidCap 400 Revenue ...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-08	20	5.5%	0.39	0.614
REM	iShares Mortgage Real Estate ETF	Real Estate	cluster_2	LONG	2026-04-09	19	2.7%	0.40	0.600
IYE	iShares U.S. Energy ETF	Energy	cluster_16	LONG	2026-05-01	3	-3.1%	0.02	0.588
IYJ	iShares U.S. Industrials ETF	Industrials	cluster_2	LONG	2026-04-10	18	2.7%	0.43	0.575
SDVY	First Trust SMID Cap Rising Div...	Dividend Equity	cluster_2	LONG	2026-04-09	19	2.2%	0.43	0.569
XLE	State Street Energy Select Sect...	Energy	cluster_16	LONG	2026-05-01	3	-3.1%	0.02	0.558
HYLS	First Trust Tactical High Yield...	High Yield Bond	cluster_8	LONG	2026-04-08	20	0.7%	0.44	0.555
IXC	iShares Global Energy ETF	Energy	cluster_16	LONG	2026-05-01	3	-2.8%	0.02	0.555
FTXR	First Trust Nasdaq Transportati...	Industrials	cluster_2	LONG	2026-04-13	17	5.1%	0.48	0.521
IDU	iShares U.S. Utilities ETF	Utilities	cluster_19	LONG	2026-04-29	5	-0.2%	0.39	0.485
SHYG	iShares 0-5 Year High Yield Cor...	High Yield Bond	cluster_8	LONG	2026-04-07	21	1.1%	0.53	0.466
VRP	Invesco Variable Rate Preferred...	Preferred Stock	cluster_8	LONG	2026-04-09	19	0.6%	0.55	0.451
PKW	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	cluster_2	LONG	2026-04-09	19	1.8%	0.55	0.450
ACWV	iShares MSCI Global Min Vol Fac...	Global Equity	cluster_5	LONG	2026-04-30	4	0.1%	0.56	0.433
XSLV	Invesco S&P SmallCap Low Volati...	U.S. Small Cap Low ...	cluster_2	LONG	2026-04-08	20	1.8%	0.61	0.395
SJNK	State Street SPDR Bloomberg Sho...	High Yield Bond	cluster_8	LONG	2026-04-07	21	1.1%	0.62	0.384
IGE	iShares North American Natural ...	Energy	cluster_11	FLAT	2026-05-01	3	-0.8%	0.19	0.362
FTA	First Trust Large Cap Value Alp...	Value Equity	cluster_5	LONG	2026-04-08	20	0.9%	0.65	0.349
IYG	iShares U.S. Financial Services...	Financials	cluster_12	FLAT	2026-04-08	20	2.0%	0.66	0.343
HYD	VanEck High Yield Muni ETF	Municipal Bond	cluster_18	FLAT	2026-04-08	20	0.8%	0.67	0.331

FXO	First Trust Financials AlphaDEX	Financials	cluster_12	FLAT	2026-04-09	19	1.9%	0.67	0.328
KWEB	KraneShares CSI China Internet ...	China Equity	cluster_15	LONG	2026-05-06	0	0.0%	0.54	0.297
KBE	State Street SPDR S&P Bank ETF	Financials	cluster_12	FLAT	2026-04-07	21	6.2%	0.70	0.295
EWJV	iShares MSCI Japan Value ETF	Japan Equity	cluster_22	FLAT	2026-05-04	2	3.8%	0.63	0.260
IAT	iShares U.S. Regional Banks ETF	Financials	cluster_12	FLAT	2026-04-08	20	2.9%	0.75	0.253
KBWB	Invesco KBW Bank ETF	Financials	cluster_12	FLAT	2026-04-07	21	7.4%	0.75	0.247
BNDX	Vanguard Total International Bo...	International Fixed...	cluster_4	FLAT	2026-05-06	0	0.0%	0.74	0.208
FTCS	First Trust Capital Strength ETF	U.S. Quality / Fact...	cluster_5	LONG	2026-05-04	2	0.4%	0.76	0.204
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected...	cluster_21	LONG	2026-05-06	0	0.0%	0.83	0.114
JETS	U.S. Global Jets ETF	Industrials	cluster_2	LONG	2026-05-06	0	0.0%	0.76	0.107
MOO	VanEck Agribusiness ETF	Consumer Staples	cluster_11	FLAT	2026-05-06	0	0.0%	0.51	0.101
IAGG	iShares International Aggregate...	International Fixed...	cluster_4	FLAT	2026-05-06	0	0.0%	0.83	0.074
PPH	VanEck Pharmaceutical ETF	Healthcare	cluster_10	FLAT	2026-05-06	0	0.0%	0.62	0.068
SLV	iShares Silver Trust	Silver	cluster_13	FLAT	2026-05-06	0	0.0%	0.84	-0.027
SIVR	abrdn Physical Silver Shares ETF	Silver	cluster_13	FLAT	2026-05-06	0	0.0%	0.84	-0.027
BAB	Invesco Taxable Municipal Bond ...	Taxable Municipal B...	cluster_4	FLAT	2026-05-06	0	0.0%	0.91	-0.028
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_18	FLAT	2026-05-06	0	0.0%	0.91	-0.031
TOTL	State Street DoubleLine Total R...	Active Fixed Income	cluster_4	FLAT	2026-05-06	0	0.0%	0.97	-0.095
XHE	State Street SPDR S&P Health Ca...	Healthcare	cluster_10	FLAT	2026-05-06	0	0.0%	0.87	-0.110
IUSB	iShares Core Universal USD Bond...	Aggregate Fixed Inc...	cluster_4	FLAT	2026-05-06	0	0.0%	0.97	-0.137
GNMA	iShares GNMA Bond ETF	Mortgage-Backed Fix...	cluster_4	FLAT	2026-05-06	0	0.0%	0.98	-0.167
AGGY	WisdomTree Yield Enhanced U.S. ...	Aggregate Fixed Inc...	cluster_4	FLAT	2026-05-06	0	0.0%	0.97	-0.181
GVI	iShares Intermediate Government...	Fixed Income	cluster_4	FLAT	2026-05-06	0	0.0%	0.98	-0.188
FBND	Fidelity Total Bond ETF	Active Fixed Income	cluster_4	FLAT	2026-05-06	0	0.0%	0.98	-0.212
BOND	PIMCO Active Bond Exchange-Trad...	Active Fixed Income	cluster_4	FLAT	2026-05-06	0	0.0%	0.99	-0.220
PHO	Invesco Water Resources ETF	Utilities	cluster_2	LONG	2026-05-06	0	0.0%	0.98	-0.232
IBB	iShares Biotechnology ETF	Healthcare	cluster_10	FLAT	2026-05-06	0	0.0%	0.92	-0.241
IGLB	iShares 10 Year Investment Grad...	Investment Grade Co...	cluster_4	FLAT	2026-05-06	0	0.0%	0.99	-0.250
COPX	Global X Copper Miners ETF	Materials	cluster_11	FLAT	2026-05-06	0	0.0%	0.97	-0.273
QLTA	iShares Aaa A Rated Corporate B...	Investment Grade Co...	cluster_4	FLAT	2026-05-06	0	0.0%	1.00	-0.290
VMBS	Vanguard Mortgage-Backed Securi...	Mortgage-Backed Fix...	cluster_4	FLAT	2026-05-06	0	0.0%	0.98	-0.296
LQD	iShares iBoxx \$ Investment Grad...	Investment Grade Co...	cluster_4	FLAT	2026-05-06	0	0.0%	0.99	-0.297
VCLT	Vanguard Long-Term Corporate Bo...	Investment Grade Co...	cluster_4	FLAT	2026-05-06	0	0.0%	0.99	-0.320
AGG	iShares Core U.S. Aggregate Bon...	Aggregate Fixed Inc...	cluster_4	FLAT	2026-05-06	0	0.0%	0.99	-0.324
SPLB	State Street SPDR Portfolio Lon...	Investment Grade Co...	cluster_4	FLAT	2026-05-06	0	0.0%	1.00	-0.342
AGZ	iShares Agency Bond ETF	Agency Fixed Income	cluster_4	FLAT	2026-05-06	0	0.0%	0.98	-0.378
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_15	LONG	2026-05-04	2	0.6%	0.92	-0.467
GUNR	FlexShares Global Upstream Natu...	Materials	cluster_11	FLAT	2026-05-05	1	-0.1%	0.90	-0.502
RSX			cluster_14	FLAT	2025-12-22	92	0.0%		

Today's Crossovers

Buy crossovers

level	id	name	category	cluster	cluster_signal	spread_buy	prob_buy	nss
etf	KWEB	KraneShares CSI China Internet ...	China Equity	cluster_15	LONG	0.100	0.84	0.297
etf	BNDX	Vanguard Total International Bo...	International Fixed...	cluster_4	FLAT	0.068	0.95	0.208
etf	LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected...	cluster_21	LONG	0.101	0.94	0.114
etf	JETS	U.S. Global Jets ETF	Industrials	cluster_2	LONG	0.383	0.87	0.107
etf	MOO	VanEck Agribusiness ETF	Consumer Staples	cluster_11	FLAT	0.410	0.61	0.101
etf	IAGG	iShares International Aggregate...	International Fixed...	cluster_4	FLAT	0.052	0.90	0.074
etf	PPH	VanEck Pharmaceutical ETF	Healthcare	cluster_10	FLAT	0.123	0.69	0.068
etf	SLV	iShares Silver Trust	Silver	cluster_13	FLAT	0.106	0.82	-0.027
etf	SIVR	abrdn Physical Silver Shares ETF	Silver	cluster_13	FLAT	0.114	0.82	-0.027
etf	BAB	Invesco Taxable Municipal Bond ...	Taxable Municipal B...	cluster_4	FLAT	0.016	0.88	-0.028
etf	SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_18	FLAT	0.006	0.88	-0.031
etf	TOTL	State Street DoubleLine Total R...	Active Fixed Income	cluster_4	FLAT	0.026	0.87	-0.095
etf	XHE	State Street SPDR S&P Health Ca...	Healthcare	cluster_10	FLAT	0.072	0.76	-0.110
etf	IUSB	iShares Core Universal USD Bond...	Aggregate Fixed Inc...	cluster_4	FLAT	0.042	0.84	-0.137
etf	GNMA	iShares GNMA Bond ETF	Mortgage-Backed Fix...	cluster_4	FLAT	0.040	0.81	-0.167
etf	AGGY	WisdomTree Yield Enhanced U.S. ...	Aggregate Fixed Inc...	cluster_4	FLAT	0.034	0.79	-0.181
etf	GVI	iShares Intermediate Government...	Fixed Income	cluster_4	FLAT	0.042	0.79	-0.188
etf	FBND	Fidelity Total Bond ETF	Active Fixed Income	cluster_4	FLAT	0.022	0.77	-0.212
etf	BOND	PIMCO Active Bond Exchange-Trad...	Active Fixed Income	cluster_4	FLAT	0.050	0.77	-0.220
etf	PHO	Invesco Water Resources ETF	Utilities	cluster_2	LONG	0.130	0.75	-0.232
etf	IBB	iShares Biotechnology ETF	Healthcare	cluster_10	FLAT	0.940	0.68	-0.241
etf	IGLB	iShares 10 Year Investment Grad...	Investment Grade Co...	cluster_4	FLAT	0.063	0.74	-0.250
etf	COPX	Global X Copper Miners ETF	Materials	cluster_11	FLAT	0.355	0.70	-0.273
etf	QLTA	iShares Aaa A Rated Corporate B...	Investment Grade Co...	cluster_4	FLAT	0.032	0.71	-0.290
etf	VMBS	Vanguard Mortgage-Backed Securi...	Mortgage-Backed Fix...	cluster_4	FLAT	0.020	0.68	-0.296
etf	LQD	iShares iBoxx \$ Investment Grad...	Investment Grade Co...	cluster_4	FLAT	0.079	0.70	-0.297
etf	VCLT	Vanguard Long-Term Corporate Bo...	Investment Grade Co...	cluster_4	FLAT	0.066	0.68	-0.320
etf	AGG	iShares Core U.S. Aggregate Bon...	Aggregate Fixed Inc...	cluster_4	FLAT	0.002	0.66	-0.324
etf	SPLB	State Street SPDR Portfolio Lon...	Investment Grade Co...	cluster_4	FLAT	0.016	0.65	-0.342
etf	AGZ	iShares Agency Bond ETF	Agency Fixed Income	cluster_4	FLAT	0.005	0.60	-0.378

Sell crossovers

level	id	name	category	cluster	cluster_signal	spread_sell	prob_sell	nss
etf	UUP	Invesco DB USD Index Bullish Fu...	Currency	cluster_27	LONG	-0.011	0.57	-0.557

etf	FNCL	Fidelity MSCI Financials Index ...	Financials	cluster_12	FLAT	-0.065	0.74	0.257
etf	XLF	State Street Financial Select S...	Financials	cluster_12	FLAT	-0.046	0.74	0.260
etf	VFH	Vanguard Financials ETF	Financials	cluster_12	FLAT	-0.056	0.72	0.280

Net Signal Strength (NSS)

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

Top 20 most bullish (NSS desc)

level	id	name	category	cluster	cluster_signal	prob_buy	prob_sell	nss
etf	QTEC	First Trust NASDAQ-100-Technolo...	Technology	cluster_1	LONG	1	0	1
etf	XSD	State Street SPDR S&P Semicondu...	Technology	cluster_1	LONG	1	0	1
etf	VLUE	iShares MSCI USA Value Factor E...	U.S. Value Factor E...	cluster_2	LONG	1	0	1
etf	SCHD	Schwab US Dividend Equity ETF	U.S. Dividend Equity	cluster_5	LONG	1	0	1
etf	MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short ...	cluster_24	FLAT	1	0	1
etf	ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short ...	cluster_24	FLAT	1	0	1
etf	SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short ...	cluster_24	FLAT	1	0	1
etf	BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short ...	cluster_24	FLAT	1	0	1
etf	GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short ...	cluster_24	FLAT	1	0	1
etf	FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed...	cluster_23	LONG	1	0	1
etf	IBDR	iShares iBonds Dec 2026 Term Co...	Target Maturity Bond	cluster_4	FLAT	1	0	1
etf	FTXL	First Trust Nasdaq Semiconducto...	Technology	cluster_1	LONG	1	0	1
etf	PTNQ	Pacer Trendpilot 100 ETF	Trend-Following Equ...	cluster_1	LONG	1	0	1
etf	NXTG	First Trust Indxx NextG ETF	Technology	cluster_1	LONG	1	0	1
etf	EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_7	LONG	1	0	1
etf	FLRN	State Street SPDR Bloomberg Inv...	Floating Rate Fixed...	cluster_23	LONG	1	0	1
etf	FLOT	iShares Floating Rate Bond ETF	Floating Rate Fixed...	cluster_23	LONG	1	0	1
etf	PSI	Invesco Semiconductors ETF	Technology	cluster_1	LONG	1	0	1
etf	ICLN	iShares Global Clean Energy ETF	Energy	cluster_25	LONG	1	0	1
etf	NULV	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	cluster_5	LONG	1	0	1

Top 20 most bearish (NSS asc)

level	id	name	category	cluster	cluster_signal	prob_buy	prob_sell	nss
etf	IHI	iShares U.S. Medical Devices ETF	Healthcare	cluster_10	FLAT	0.00	1.00	-1.000
etf	DZZ	DB Gold Double Short ETN due Fe...	Leveraged / Inverse...	cluster_27	LONG	0.00	1.00	-1.000
etf	PSQ	ProShares Short QQQ	Leveraged / Inverse	cluster_9	FLAT	0.00	1.00	-1.000
etf	QID	ProShares UltraShort QQQ	Leveraged / Inverse	cluster_9	FLAT	0.00	1.00	-1.000
etf	SQQQ	ProShares UltraPro Short QQQ	Leveraged / Inverse	cluster_9	FLAT	0.00	1.00	-0.999
etf	SCO	ProShares UltraShort Bloomberg ...	Leveraged / Inverse...	cluster_9	FLAT	0.00	1.00	-0.999
etf	EDV	Vanguard Extended Duration Trea...	Fixed Income	cluster_4	FLAT	0.00	1.00	-0.995

etf	GOAU	US Global GO Gold and Precious ...	Materials	cluster_13	FLAT	0.01	1.00	-0.992
etf	SH	ProShares Short S&P500	Leveraged / Inverse	cluster_9	FLAT	0.00	0.99	-0.991
etf	SDS	ProShares UltraShort S&P500	Leveraged / Inverse	cluster_9	FLAT	0.00	0.99	-0.988
etf	EUM	ProShares Short MSCI Emerging M...	Leveraged / Inverse	cluster_9	FLAT	0.00	0.99	-0.985
etf	RWM	ProShares Short Russell2000	Leveraged / Inverse	cluster_9	FLAT	0.00	0.98	-0.984
etf	SPXS	Direxion Daily S&P 500 Bear 3X ...	Leveraged / Inverse	cluster_9	FLAT	0.00	0.98	-0.983
etf	SPXU	ProShares UltraPro Short S&P500	Leveraged / Inverse	cluster_9	FLAT	0.00	0.98	-0.983
etf	NUGT	Direxion Daily Gold Miners Inde...	Leveraged / Inverse...	cluster_13	FLAT	0.02	1.00	-0.980
cluster	cluster_9	Leveraged / Inverse (16/22)	Leveraged / Inverse			0.00	0.98	-0.980
etf	EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_14	FLAT	0.02	1.00	-0.976
etf	TWM	ProShares UltraShort Russell2000	Leveraged / Inverse	cluster_9	FLAT	0.00	0.98	-0.975
etf	GDX	VanEck Gold Miners ETF	Materials	cluster_13	FLAT	0.03	1.00	-0.974
etf	TLT	iShares 20+ Year Treasury Bond ...	Fixed Income	cluster_4	FLAT	0.03	1.00	-0.970

Run Parameters

	parameter	value
short_ma	short_ma	5
buy_ma	buy_ma	35
sell_ma	sell_ma	20
vol_window	vol_window	20
horizon	horizon	5
cluster_smoothing_window	cluster_smoothing_window	3